Coral Reefs (2024) 43:1857–1877
https://doi.org/10.1007/s00338-024-02569-7

NOTE

The mitochondrial genomes of the reef-dwelling spiny lobsters *Panulirus echinatus* and *Panulirus interruptus* with insights into the phylogeny and adaptive evolution of protein-coding genes in the Achelata

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Received: 27 November 2023 / Accepted: 9 September 2024 / Published online: 21 October 2024

Abstract Temperature and oxygen levels drive the evolution of morphological, behavioral, and physiological traits in marine invertebrates, including crustaceans. Environmental conditions are also expected to prompt the adaptive evolution of mitochondrial protein-coding genes (PCGs), which are vital for energy production via the oxidative phosphorylation pathway. We formally tested for adaptive evolution in mitochondrial protein-coding genes in representatives of the decapod infraorder Achelata, including two spiny lobsters, *Panulirus echinatus* and *P. interruptus*, for which we sequenced complete mitochondrial genomes (15,644 and 15,659 bp long, respectively). A phylomitogenomic analysis supported the monophyly of the genus *Panulirus*, the families Palinuridae and Scyllaridae, and the infraorder Achelata. Over the strong negative selection background observed for mitochondrial PCGs in the Achelata, signatures of positive selective pressure were detected within PCGs in equatorial *Panulirus* spp. and deepwater Scyllaridae. In *Panulirus* spp. inhabiting equatorial latitudes with consistently high temperatures, the Datamonkey analysis RELAX suggested intensified purifying selection strength in 9 of the 13 PCGs and relaxation in purifying selection strength in *atp6*, while aBSREL, BUSTED, and MEME recovered signatures of positive selection on PCGs within Complex I, III, and IV PCGs. Likewise, in Scyllaridae species inhabiting depths with low-oxygen levels, RELAX indicated relaxed selection strength in 6 of the 13 PCGs, while aBSREL, BUSTED, and MEME recovered signatures of positive selection on PCGs within Complexes I, III, IV, and V. The newly assembled mitochondrial genomes of *P. echinatus* and *P. interruptus* represent new genomic resources to aid with the conservation and management of lobsters targeted by major fisheries and contribute to our understanding of how environmental conditions drive adaptive evolution in spiny and slipper lobster mitochondrial PCGs.

Resumo A temperatura e os níveis de oxigênio determinam a evolução das caraterísticas morfológicas, comportamentais e fisiológicas dos invertebrados marinhos, incluindo os crustáceos. As condições ambientais também devem induzir a evolução adaptativa dos genes codificadores de proteínas mitocondriais (PCGs), vitais para a produção de energia através da via de fosforilação oxidativa. Testamos a evolução adaptativa dos genes mitocondriais codificadores de proteínas em representantes dos decápodes, infraordem Achelata, incluindo duas espécies de lagostas, *Panulirus echinatus* e *P. interruptus*, para as quais sequenciamos genomas mitocondriais completos (15 644 e 15 659 pb de comprimento, respetivamente). Uma análise filogenômica apoiou a monofilia do gênero *Panulirus*, das famílias Palinuridae e Scyllaridae, e da infraordem Achelata. Sob forte processo de seleção negativa observado para PCGs mitocondriais em Achelata, foram detectados sinais de pressão seletiva positiva em PCGs de *Panulirus* spp. e Scyllaridae de águas profundas. Em *Panuli-*

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/s00338-024-02569-7.

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rus spp. que habitam latitudes equatoriais com temperaturas consistentemente elevadas, a análise RELAX (Datamonkey web server) sugeriu uma intensificação da força de seleção purificadora em 9 dos 13 PCGs e um relaxamento da força de seleção purificadora em *atp6*, enquanto os métodos aBSREL, BUSTED, e MEME recuperaram assinaturas de seleção positiva em PCGs, dentro dos Complexos I, III e IV. Da mesma forma, em espécies de Scyllaridae que habitam profundidades com baixos níveis de oxigênio, RELAX indicou um relaxamento da força de seleção em 6 dos 13 PCGs, enquanto aBSREL, BUSTED e MEME recuperaram assinaturas de seleção positiva em PCGs para os Complexos I, III, IV e V. Os genomas mitocondriais recentemente montados de *P. echinatus* e *P. interruptus* representam novos recursos genômicos para auxiliar na conservação e gestão pesqueira destas espécies de lagostas, e contribuem para a nossa compreensão de como as condições ambientais impulsionam a evolução adaptativa nos genes mitocôndrias codificadores de proteínas em lagostas espinhosas e sapateiras.

Keywords Achelata · Mitochondrial genome · Crawfish · Lobsters · Selective pressure · Phylomitogenomics

Introduction

The mitochondrial genomes of vertebrates and invertebrates, including marine decapod crustaceans, are most often circular molecules 15,000–16,000 bp in length that encode 13 protein-coding genes (PCGs), 22 transfer RNA genes, 2 ribosomal RNA genes, and a non-coding control region or D-loop (Bernt et al. 2013). Mitochondrial PCGs encode several components of the oxidative phosphorylation (OXPHOS) pathway within Complex I (*nad1, nad2, nad3, nad4, nad4L, nad5,* and *nad6*), Complex III (*cob*), Complex IV (*cox1, cox2*, and *cox3*), and Complex V (*atp6* and *atp8*) (Luo et al. 2013; Chaban et al. 2014). In the OXPHOS pathway, the electron transport chain (Complexes I-IV) first pumps protons into the mitochondrial intermembrane space (Luo et al. 2013; Chaban et al. 2014). Protons enter the electron transport chain through either NADH dehydrogenase (Complex I) binding of NADH or nuclear-coded succinate dehydrogenase (Complex II) oxidation of succinate (Luo et al. 2013; Chaban et al. 2014). Protons are then pumped into the intermembrane space via electron transference, where cytochrome c oxidoreductase (Complex III) oxidizes ubiquinone to ubiquinol and cytochrome c oxidase (Complex IV) binds electrons to oxygen to form water (Luo et al. 2013; Chaban et al. 2014). ATP synthase (Complex V) then produces adenosine triphosphate (ATP) through oxidative phosphorylation (Luo et al. 2013; Chaban et al. 2014). Overall, mitochondria are most relevant for energy production and storage as ATP during cellular respiration in animals (Lehninger 1964).

Empirical studies during the last decades have shown mitochondrial genomes experience a greater mutation rate than nuclear genomes, purportedly due to heightened exposure to damaging oxidation and maternal inheritance lacking mutation-countering recombination (Neiman & Taylor 2009; Palozzi et al. 2018). Thus, each mitochondrial PCG that encodes components of the OXPHOS pathway is susceptible to site-specific substitutions with the potential to alter overall protein structure or modify components tied to protein function, such as active sites (Zhang et al. 2013). Changes to protein structure or components may impact substrate binding capabilities, OXPHOS efficiency, and the production of reactive oxygen species (ROS) that interfere with cell activities (Abele et al. 2007). Subsequently, mitochondrial genomes represent a suitable model system to understand the molecular adaptation of organisms to dissimilar environments with contrasting biotic and abiotic conditions in genes that are indispensable for metabolic performance.

A limited number of studies in marine invertebrates, including crustaceans, show that under extreme temperature and low-oxygen conditions, genes in two complexes are likely exposed to positive selection (Zhou et al. 2014; Wang et al. 2017; Sun et al. 2018, 2021). The first affected complex is the NADH enzyme (Complex I), a component of the electron transport chain that functions as a proton pump (da Fonseca et al. 2008; Hassanin et al. 2009; Chaban et al. 2014). Complex I forms a proton gradient by oxidizing NADH, then pumping H+ from the mitochondrial matrix to the intermembrane space (Chaban et al. 2014). Oxidation and pump speed are impacted by temperature, with optimal Complex I function achieved within a specific range of temperatures (Iftikar et al. 2014; Ekström et al. 2017; Christen et al. 2018; Oellermann et al. 2020). In ectotherms, which cannot regulate body temperature, the environmental temperature may fluctuate outside the range for optimal enzyme function (Farrell 2002; Lannig et al. 2004; Abele et al. 2007), which has been shown to increase mitochondrial activity (Farrell 2002; Lannig et al. 2004; Abele et al. 2007; Li et al. 2013). When oxygen consumption is greater than the circulatory system can deliver, cellular hypoxia occurs (Farrell 2002; Lannig et al. 2004; Abele et al. 2007) and prompts elevated respiration to counter increased proton leakage, reduced membrane potential, and heightened production of reactive oxygen species (ROS) (Massabuau 2001; Stillman 2002; Turrens 2003; Oellermann et al. 2020; Abele et al. 2007). Left uncorrected, whole-body decline at the cellular level occurs (Massabuau 2001; Stillman 2002; Turrens 2003; Oellermann et al. 2020; Abele et al. 2007). Temperature sensitivity in ectotherm Complex I thus likely reflects environmental temperature fluctuations. While narrower optimal temperature ranges are expected in tropical species experiencing more consistent temperatures, broader tolerance ranges are expected in temperate species experiencing greater fluctuations (Stillman 2002; Iftikar et al. 2014; Oellermann et al. 2020). As such, climate-specific (temperature) sensitivity likely drives the evolution of Complex I genes to optimize proton pump operation under specific local temperature fluctuations.

The second complex likely exposed to adaptive evolution in invertebrates is the ATP synthase enzyme complex (Complex V). Complex V is the final respiratory chain complex and produces ATP by transporting electrons to perform oxidative phosphorylation on ADP (Weiss et al. 1991; Mishmar et al. 2003; Chaban et al. 2014; Zhou et al. 2014). A proton gradient generated by the electron transport chain drives H + from the intermembrane space through a proton channel to the ATP synthase enzyme (Jonckheere et al. 2018). As ATP demand increases, the movement of H + through these proton channels also increases to maintain enough energy for normal cellular activities (Devin & Rigoulet 2007). Previous research suggests that increased ATP demand, and therefore, elevated proton channel activity occurs in species that inhabit perpetually low-oxygen conditions, such as in marine crustaceans (Zhang et al. 2013; Wang et al. 2017) and high-altitude ground birds (Zhou et al. 2014). In marine systems specifically, dissolved oxygen (DO) concentration declines at depth, as average annual mid-latitude (30°N – 30°S) DO is 200 µmol/kg to 260 µmol/kg at the surface and 4 µmol/kg to 190 µmol/kg at 850 m (Garcia et al. 2018). In invertebrates inhabiting low-oxygen deep environments, enzymatic structural changes that preserve cellular function by optimizing proton channel efficiency are likely, and if present, would be reflected at the genomic level; positive selection in mitochondrial genes belonging to Complex V is observable (Zhang et al. 2013; Zhou et al. 2014; Wang et al. 2017).

Globally distributed, marine decapod crustaceans exhibit major differences in habitat use and correlated environmental conditions (Castro et al. 2015). Among decapod crustaceans, the infraorder Achelata inhabits a wide range of latitudes and depths (George 2006; Lavalli et al. 2019) and is thus an ideal system to examine adaptive evolution in mitochondrial PCGs driven by consistently high versus fluctuating moderate temperatures and low (hypoxic) vs high (normoxic) oxygen environmental conditions. The Achelata, with approximately 150 extant species (Chan 2010; Davis et al. 2015), contains exclusively marine spiny (fam. Palinuridae) and slipper (fam. Scyllaridae) lobsters characterized by long-lived, motile phyllosoma larvae with a high dispersal capacity (Holthuis 1946; Pollock 1990; George 2006). Representatives of the family Palinuridae primarily inhabit tropical and subtropical regions (35 °N to 35 °S) at known depths from ~ 0 m to ~ 1300 m (George & Main 1967; George 2006), and the most species-rich palinurid genus, *Panulirus*, contains 20 recognized species and 5 subspecies throughout shallow tropical and subtropical oceanic waters, with known depths typically ~ 0 m to ~ 100 m but occasionally up to ~ 180 m (Holthuis et al. 1980; George 2006). In turn, slipper lobsters in the family Scyllaridae range throughout temperate, subtropical, and tropical climates (45 °N to 45 °S), with known depths extending from ~ 1 m to ~ 850 m (Holthuis 1991; Lavalli et al. 2019). Palinuridae and Scyllaridae thus inhabit environments that exhibit either consistent or fluctuating temperatures and depths with different levels of oxygen availability. Since these environmental conditions affect OXPHOS efficiency, it is possible to test for adaptive evolution (positive selection) on Achelata PCGs, specifically in *nad2, nad4,* and *nad6* in Complex I, and *atp6* and *atp8* in Complex V. Overall, the Achelata is an interesting clade to evaluate how temperature and depth drive mitochondrial genomic adaptations.

This study tested for signatures of positive selection in protein-coding genes encoded by the mitochondrial genome using representatives of the Achelata as a model system. We first assembled and described the mitogenomes of *P. echinatus* and *P. interruptus* from next-generation sequencing reads. We then explored the phylogenetic position of the studied species within the Achelata based on PCGs and generated a maximum likelihood (ML) phylogenetic hypothesis for all representatives of the Achelata with mitochondrial genomes assembled and available in NCBI's GenBank database. Finally, selective pressure analyses were conducted on protein-coding genes encoded in the mitochondrial genome. KaKs Calculator v.2.0 (Wang et al. 2009) initially evaluated the prevailing direction and strength of selection on *P. echinatus* and *P. interruptus* mitochondrial PCGs. The molecular basis for patterns of adaptive evolution in the two Achelata lineages experiencing disparate environmental conditions was then further explored using branch (RELAX) (Wertheim et al. 2015), branch-site (aBSREL, BUSTED) (Murrell et al. 2015; Smith et al. 2015), and site (MEME, FUBAR, SLAC) (Murrell et al. 2012, 2013; Pond & Frost 2005b) models in the webserver Datamonkey (Pond & Frost 2005a; Poon et al. 2009; Delport et al. 2010; Weaver et al. 2018). First, we hypothesized that *Panulirus* spiny lobsters inhabiting tropical, high temperature water (above 20 °C year-round) in equatorial regions experience stronger positive selection on *nad2*, *nad4*, and *nad6* than in other Achelata, as Complex I is refined to operate within an optimal range of high temperatures and tropical regions have high and the narrowest temperature fluctuations. We additionally hypothesized that compared to other Achelata, Scyllaridae with known depth ranges over 200 m experience positive selection on *atp6* and *atp8*, as comparatively low-oxygen waters require elevated Complex V activity to maintain the mitochondrial membrane proton gradient.

Methods

Collection and Sequencing of Mitochondrial Genomes

The mitochondrial genome of *P. echinatus* was sequenced from one specimen caught and euthanized by fishermen near Ponta Grossa Beach (4°37′48.0" S, 37°30′22.9" W), Ceará, Brazil. The specimen was transported to the Northeast Biotechnology Network RENORBIO laboratory, Universidade Federal do Piauí, Teresina, Piauí, Brazil. The mitochondrial genome of *P. interruptus* was sequenced from one adult individual captured and euthanized by the Centro de Investigación Científica y de Educación Superior de Ensenada (CICESE) aquaculture facility (31°52’03.40" N, -116°39'54.18" W), Ensenada, Baja California, México. The specimen was transported to the Laboratorio de Organismos Acuáticos, Instituto de Ciencias del Mar y Limnología (ICMyL), Universidad Autónoma de México (UNAM). Pereiopod muscle was extracted from both specimens with forceps, and the tissue was immediately preserved in 95% ethyl alcohol. Total gDNA was extracted from muscle tissue using an EZNA Genomic DNA Purification kit (Omega Bio-Tek, Norcross, GA). Genomic DNA isolation was performed via a phenol:chloroform:isoamyl alcohol (25:24:1) extraction of the SDS/proteinase-K-digested tissue, followed by ethanol precipitation (Sambrook et al. 1989). The *P. echinatus* DNA sample was then shipped to the Marine Gene Probe Laboratory (MGPL), Dalhousie University, Canada. The *P. interruptus* sample was shipped to Georgia Genomics and Bioinformatics Core (GGBC), University of Georgia, Athens, GA, USA, for library preparation and Illumina paired-end shotgun sequencing.

At MGPL, the *P. echinatus* mitochondrial genome was sequenced after preparing an Illumina paired-end library using 1 ng DNA following the standard protocol of the Illumina Nextera XT Library Preparation kit (Illumina, San Diego, USA). DNA was then tagged and fragmented by the Nextera XT transposome, followed by PCR amplification, AMPure XP magnetic bead purification (Agencourt Bioscience, Beverly, USA) and the Illumina Nextera XT bead-based normalization protocol. The DNA library was finally sequenced with the MiSeq Benchtop Sequencer (Illumina, San Diego, USA) using a 2×500 cycle to produce 250 bp paired-end reads (PE). The *P. interruptus* mitochondrial genome was sequenced at GGBC. Illumina® libraries were first prepared by shearing ~ 1 µg of gDNA (using a Covaris instrument), then following the standard protocol of the Illumina Truseq DNA Library Preparation kit using a multiplex identifier adaptor index (Illumina). The DNA library was finally sequenced with the MiSeq v2® platform using a 2×500 cycle to produce 250 bp paired-end reads.

Mitochondrial Genome Assembly of *Panulirus echinatus* and *Panulirus interruptus*

The mitochondrial genomes of *P. echinatus* and *P. interruptus* were de novo-assembled using the pipeline GetOrganelle v1.6.4 (Jin et al. 2020). The mitochondrial genome of the congeneric *P. argus*, available in GenBank (NC_039671—Baeza 2018), was used as a seed. The run used kmer sizes of 21, 55, 85, and 115. The reads were not filtered before assembly, following the developer's suggestions.

Mitochondrial Genome Annotation and Analysis

The newly assembled mitochondrial genomes belonging to *P. echinatus* and *P. interruptus* were first annotated using the web server MITOS (http://mitos.bioinf.uni-leipzig.de) and MITOS2 (http://mitos2.bioinf.uni-leipzig.de) (Bernt et al. 2013; Donath et al. 2019). Next, the in silico protein sequence annotations were manually curated, with a focus on protein-coding gene (PCGs) start and stop codons corrections, using the ExPASy Translate tool in the web server Expasy (https://web.expasy.org) (Artimo et al. 2012) and the software MEGAX (Kumar et al. 2018). Visualizations of the genomes were performed in the web server GenomeVx (http://wolfe.ucd.ie/GenomeVx/) (Conant & Wolfe 2008).

The nucleotide composition of the entire mitochondrial chromosome was estimated in the software MEGAX (Kumar et al. 2018). Codon usage for PCGs in the mitochondrial genomes was analyzed with Codon Usage web server Sequence Manipulation Suite v.2 (https://www.bioinformatics.org/sms2/codon_usage.html) (Stothard 2000). Relative synonym codon usage (RSCU) of PCGs in the positive and negative strands was also determined in EZcodon (http://ezmito.unisi.it/ezcodon) (Lee 2018) within EZmito (Cucini et al. 2021).

The secondary structure of tRNA was estimated in MiTFi (Jühling et al. 2012) as implemented in MITOS (Bernt et al. 2013) and depicted using the web server Forna (http://rna.tbi.univie.ac.at/forna/)) (Kerpedjiev et al. 2015).

The control region was analyzed for repetitive elements using Microsatellite Repeats Finder (http://insilico.ehu.es/mini_tools/microsatellites/) (Joseba 2012). Microsatellite repeats were examined with Tandem Repeats Finder (https://tandem.bu.edu/trf/trf.basic.submit.html) (Benson 1999). Control region secondary structures were evaluated using the Predict a Secondary Structure web server on the RNAstructure website (https://rna.urmc.rochester.edu/RNAstructureWeb/Servers/Predict1/Predict1.html) (Reuter & Mathews 2010) with default parameters.

Phylomitogenomics of the Achelata

We analyzed the phylogenetic position of *P. echinatus* and *P. interruptus* in the infraorder Achelata based on all 13 translated PCGs. The newly assembled mitogenomes and complete mitochondrial genomes for other 15 species belonging to the family Palinuridae plus 7 species belonging to the family Scyllaridae were selected for ML phylogenetic reconstruction using the pipeline MitoPhAST v.3.0 (Tan et al. 2015). Outgroups included a member of the infraorder Caridea (*Alpheus japonicus*) and the infraorder Nephropidae (*Nephrops norvegicus*). Sequence datasets for each of the 13 PCGs were aligned and trimmed using MUSCLE Sequence Comparison by Log-Expectation (Edgar 2004a, b) in MEGA X (Kumar et al. 2018). Datasets for all 13 PCGs were then concatenated in Concatenator v.0.2.1 (Price et al. 2010; Dunn 2020; Vences et al. 2021; 2022). To determine the most suitable substitution model, each alignment was analyzed with a hierarchical hypothesis-testing framework in jModelTest2 (Darriba et al. 2012) as implemented in the server CIPRES v.3.3 (https://www.phylo.org/index.php/) (Miller et al. 2010). Finally, a maximum likelihood phylogeny was constructed from the concatenated and partitioned sequences using IQ-TREE v.2.1.2 (http://iqtree.cibiv.univie.ac.at) (Nguyen et al. 2015) with 1,000 bootstrap replicates.

Adaptive evolution of protein-coding genes

We examined the effect of two specific environments and correlated ecological conditions on the adaptive evolution of all PCGs in the Achelata. First, we tested for the effect of temperature on the adaptive evolution (positive selection) of Complex I (*nad2*, *nad4*, and *nad6*) genes in the genus *Panulirus*. We predicted that spiny lobster species inhabiting equatorial latitudes with consistent warm temperatures would experience adaptive evolution in *nad2*, *nad4*, and *nad6* mitochondrial genes compared to related species inhabiting lower latitudes and that experience less consistent temperatures during their lifetime. Next, we tested for the effect of oxygen limitation on the adaptive evolution (positive selection) of the *atp6* and *atp8* genes in Complex V in the family Scyllaridae. We predicted that slipper lobsters belonging to the family Scyllaridae that live in deep environments characterized by low-oxygen levels (including hypoxia) exhibit signatures of positive selection in mitochondrial genes *atp6* and *atp8* compared to other Achelata inhabiting more shallow environments.

To test these adaptive evolution hypotheses, we evaluated signatures of adaptive evolution on all 13 protein-coding genes. At the molecular level, adaptive evolution affects the prevalence of alleles in population gene pools by constraining (purifying or negative selection), accumulating (neutral selection), or retaining (diversifying or positive selection) amino acid substitutions. We evaluated whether prevailing positive, negative, or neutral selection occurred across each of the newly sequenced *P. echinatus* and *P. interruptus* PCGs using interspecies comparisons of average nucleotide and site substitutions in the program KaKs Calculator v.2.0 (Wang et al. 2009). However, KaKs Calculator v.2.0 only produces ω ratios averaged over all sites in a PCG and thus is a sub-optimal tool for detecting episodic positive selection, and does not specify which sites are under selection, nor compare adaptive evolution patterns within or between lineages (Wang et al. 2009). To provide a more robust description of selection strength (relaxed vs. intensified) and episodic positive selection on sites and branches across the *Panulirus* and Scyllaridae lineages, we further explored the molecular basis for adaptive evolution patterns in the genus *Panulirus* and family Scyllaridae using branch, branch-site, and site models in the web server Datamonkey (Pond & Frost 2005a; Poon et al. 2009; Delport et al. 2010; Weaver et al. 2018). We utilized the branch model to detect whether selection strength intensified or relaxed along the lineages representing the genus *Panulirus* or the family Scyllaridae. Branch-site models were used to detect signatures of positive selection across the *Panulirus* or Scyllaridae lineages and identify which sites within these lineages are likely experiencing positive selection. Lastly, we further explored the adaptive evolution of PCGs in the Achelata using site models to detect episodic positive selection on specific sites and to identify the number of sites under positive or negative selection. Additional details regarding the use and function of KaKS Calculator v.2.0 and analyses in Datamonkey are outlined below.

The program KaKs Calculator v.2.0 (Wang et al. 2009) was used to test for overall selection strength and direction across individual PCGs. KaKs Calculator v.2.0 estimates K_A (nonsynonymous substitutions per nonsynonymous site: $K_A = d_N = S_A/L_A$), K_S (synonymous substitutions per synonymous site: $K_S = d_S = S_S/L_S$), and ω (K_A / K_S). The different estimated values were based on a pairwise comparison with the related *P. argus* (NC_039671) after aligning each PCG using the program MUSCLE (Edgar 2004a, b), as implemented in MEGAX. The model γ-MYN (Wang et al. 2009) was used to account for variable mutation rates across sites. Diversifying (positive) selection on PCGs was indicated when ω (K_A / K_S) > 1, while purifying (negative) selection was indicated when ω < 1 (Wang et al. 2009).We also checked for signatures of positive selection in all 13 mitochondrial PCGs potentially driven for the colonization of/adaptation to habitats with dissimilar temperature and oxygen availability at depth in members of the Achelata using the package HyPhy (Pond & Muse 2005) in the web server Datamonkey (Pond & Frost 2005a; Poon et al. 2009; Delport et al. 2010; Weaver et al. 2018).

One branch model (BM; RELAX) (Wertheim et al. 2015), two branch-site models (BSMs; aBSREL and BUSTED) (Murrell et al. 2015; Smith et al. 2015), and three site models (SMs; MEME, FUBAR, and SLAC) (Murrell et al. 2012, 2013; Pond & Frost 2005b). were executed for each aligned PCG dataset belonging to 15 species of Palinuridae and 7 species of Scyllaridae (plus the outgroup containing 2 species) inhabiting regions of variable temperatures and depths.

The RELAX analysis determines whether the strength of selection has relaxed or intensified along specific branches in a phylogenetic tree (Werheim et al. 2015). For each of the 13 PCG datasets, neighbor-joining phylogenetic trees were first generated using the General Time-Reversible (GTR) substitution model. Next, test (= foreground) branches were assigned according to the environmental condition of interest (leading to *Panulirus* spp. inhabiting tropical equatorial regions or Scyllaridae in low-oxygen depths over 200 m) and reference (= background) branches were assigned to all other taxa. Then, each dataset was fitted with a null model (three ω classes applied across the phylogeny) and an alternative model that introduces the selection intensity parameter k and modified and fixed ω^k classes. A likelihood ratio test (LRT) was used to compare the null model to the alternative model, where $k > 1$ indicated intensified purifying selection along test branches and $k < 1$ indicated relaxation of purifying selection along test branches. Shifts in the strength of selection acting on a gene was indicated when the alternative model best fit the dataset ($p < 0.05$) (Wertheim et al. 2015). We predicted positive selection on terminal nodes (species) inhabiting either equatorial high temperatures or low-oxygen concentration at depth (foreground branches).

The aBSREL analysis assesses the proportion of sites under positive selection within selected foreground branches, i.e., those leading to *Panulirus* genera inhabiting tropical equatorial regions or Scyllaridae in low-oxygen depths over 200 m in our study. aBRSEL estimates site and branch level heterogeneity via a corrected Akaike Information Criterion (AIC_c), which infers the 'optimal' number of ω rate classes for each branch (Smith et al. 2015). Then, a full adaptive model with the 'optimal' number of ω rate classes was fitted to the dataset. An LRT performed at each branch compared the full adaptive model against a null model where branches do not experience positive selection ($\omega \leq 1$) (Smith et al. 2015). Similarly, the BUSTED analysis identifies whether a gene has undergone positive selection on at least one site and at least one foreground branch (Murrell et al. 2015). BUSTED compared the fit of an alternative (or unrestricted model) with an 'optimal' number of ω rate classes ($\omega_1 \leq \omega_2 \leq 1 \leq \omega_3$) against a null model (or constrained model) where foreground branches do not experience positive selection ($\omega_3 = 1$) (Murrell et al. 2015). "Evidence Ratios" (ER) within a threshold of 10 were also calculated for each site, providing a likelihood ratio for the alternative model representing a better fit to the dataset than the null model (Murrell et al. 2015). We predict that the alternative model would best fit our datasets if consistent warm temperatures at equatorial latitudes drives adaptive evolution in *nad2*, *nad4*, and *nad6* in the genus *Panulirus* and if low-oxygen conditions, including hypoxia, at deep habitats, drives the adaptive evolution of *atp6* and *atp8* in representatives of the family Scyllaridae.

Lastly, SM analyses were conducted only on spiny lobsters inhabiting equatorial latitudes and slipper lobsters inhabiting low-oxygen depths, as these lineages were predicted to undergo episodic positive selection. First, the Mixed Effect Model of Evolution (MEME) (Murrell et al. 2012) was used to examine branch-site episodic positive selection. MEME allows ω to vary across sites within lineages. For this test, we selected a p-value < 0.05 to indicate the probability that a site evolves under one of two ω-rate classes and minimize false-positive results. Next, a Fast Unconstrained Bayesian AppRoximation (FUBAR) analysis (Murrell et al. 2013) was conducted under default settings. FUBAR uses a Baysian approach to infer constant K_A and K_S for alignment sites and reports posterior probabilities on a scale of 0 (no selection) to 1 (positive selection) (Murrell et al. 2013). We utilized a posterior probability value of 0.95 to minimize false positive results. Lastly, a Single-Likelihood Ancestor Counting (SLAC) analysis used a combination of maximum likelihood and counting to infer constant K_A and K_S for alignment sites (Pond & Frost 2005a, b). Positive ($\omega > 1$) or negative selection ($\omega < 1$) was considered significant for sites with a p-value < 0.05. We predict positive selection on residues in *Panulirus* and Scyllaridae lineages but not in other Palinurid species if consistent warm temperatures at equatorial latitudes drives adaptive evolution in *nad2*, *nad4*, and *nad6* in *Panulirus* and if hypoxia at depth drives the adaptive evolution of *atp6* and *atp8* in Scyllaridae.

Results

Mitochondrial genomes of *Panulirus echinatus* and *Panulirus interruptus*

A total of 3,232,259 and 769,572 PE reads were generated for *P. echinatus* and *P. interruptus*, respectively, and made available in FASTQ format by the sequencing facilities. All reads are available in the short-read archive (SRA) repository (BioProject PRJNA1133224, accession numbers SRX25246072 and SRX25246073) at GeneBank. The totality of these reads was used for the mitochondrial genome assembly of *P. echinatus* and *P. interruptus*.

The pipeline GetOrganelle assembled the complete mitochondrial genomes of *P. echinatus* (GenBank accession no. MT683852) and *P. interruptus* (GenBank accession no. OR493386) with an average coverage of 12.6 × and 20.9 × per kmer and 23.1 × and 41.4 × per nucleotide, respectively. The mtDNA sequences for *P. echinatus* and *P. interruptus* were 15,644 bp and 15,659 bp in length, respectively (see Fig. 1). The newly assembled *P. echinatus* and *P. interruptus* mitogenomes encoded 13 protein-coding genes (PCGs), 22 transfer RNA genes, 2 ribosomal RNA genes (see Tables S1 and S2), and a relatively long non-coding putative Control Region located between rrnS and trnI (732 bp in length for *P. echinatus* and 724 bp for *P. interruptus*; Fig. 1). Most genes were recorded on the positive or heavy strand (H), with 4 PCGs (*nad5, nad4, nad4L*, and *nad1*), 2 rRNA (rrnaL and rrnS), and 8 tRNAs (trnF, trnH, trnP, trnL1, trnV, trnQ, trnC, and trnY) located on the negative or light strand (L). An intergenic spacer assumed to be the control region (CR), or D-loop, was 733 bp long in *P. echinatus* and 724 bp in *P. interruptus*.

Protein-coding genes in the mitochondrial genomes of *P. echinatus* and *P. interruptus* had similar start and stop codon usages. In *P. echinatus*, 8 of the 13 PCGs used ATG as the start codon (see Table 1), while *nad3* and *nad1* used ATT, and *nad2* and *nad4* used ATA as the start codons. For *cox1*, the alternative putative start codon ACG was used. Eleven PCGs terminated with conventional stop codons (TAA or TAG) (see Table 1). The incomplete stop codon T was used for *cox1* and *nad4.* Likewise, in the mitochondrial genome of *P. interruptus*, 8 of the 13 PCGs began with ATG (see Table S2), 3 used ATT, *nad1* used ATA and *cox1* used the

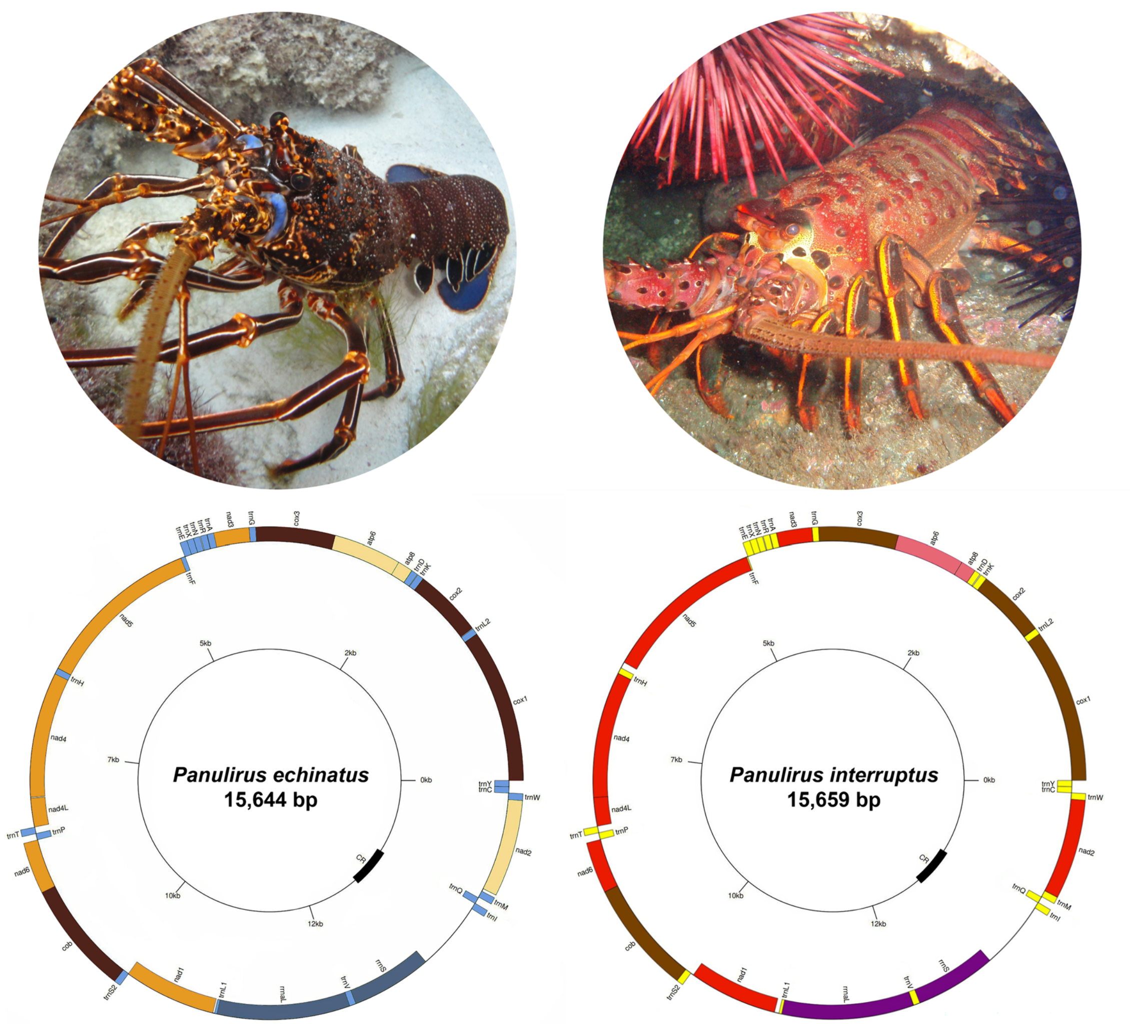

Fig. 1 Circular genome map for annotated complete mitochondrial genomes of *Panulirus echinatus* (left) and *Panulirus interruptus* (right). Included on the outer ring are 13 protein-coding genes (PCGs), 2 ribosomal RNA genes (rrnL and rrnS), and 22 transfer RNA (tRNA) genes. The unannotated control region/D-loop is noted on the inner scale ring. Photo of *P. echinatus* by Juliana de Carvalho Gaeta (2013) and *P. interruptus* by Clair Fackler (2019)

Table 1 Mitochondrial genomes of *Panulirus echinatus* and *Panulirus interruptus* with annotations of protein-coding, tRNA, rRNA, and Control Region sequences

Feature	Location (Length, bp)		Strand	Start	Stop	Anticodon	Continuity
	Panulirus echinatus[a]	*Panulirus interruptus*[b]					
cox 1	1–1534 (1534)	1–1539 (1539)	+	ACG	T*[a], TAA[b]		0[a], -5[b]
trnL2	1535–1598 (64)	1535–1600 (66)	+			TAA	6[a], 3[b]
cox2	1605–2309 (705)	1604–2308 (705)	+	ATG	TAA		-17
trnK	2293–2357 (65)	2292–2356 (65)	+			TTT	6[a], 10[b]
trnD	2364–2428 (65)	2367–2429 (63)	+			GTC	0
atp8	2429–2587 (159)	2430–2588 (159)	+	ATG	TAA		-7
atp6	2581–3258 (678)	2582–3259 (678)	+	ATG	TAA		-1
cox3	3258–4049 (792)	3259–4050 (792)	+	ATG	TAA		-2[a], -1[b]
trnG	4048–4112 (65)	4050–4114 (65)	+			TCC	0
nad3	4113–4466 (354)	4115–4468 (354)	+	ATT	TAG[a], TAA[b]		-2[a], 0[b]
trnA	4465–4530 (66)	4469–4534 (66)	+			TGC	3[a], 5[b]
trnR	4534–4597 (64)	4540–4603 (64)	+			TCG	8[a], 7[b]
trnN	4606–4670 (65)	4611–4674 (64)	+			GTT	0
trnS1	4671–4738 (68)	4675–4742 (68)	+			TCT	-1
trnE	4738–4808 (71)	4742–4812 (71)	+			TTC	0[a], 3[b]
trnF	4809–4877 (69)	4816–4881 (66)	−			GAA	−23[a], −53[b]
nad5	4855–6606 (1752)	4829–6544 (1716)	−	ATG	TAG[a], TGA[b]		0[a], 66[b]
trnH	6607–6674 (68)	6611–6675 (65)	−			GTG	−3[a], 0[b]
nad4	6672–7995 (1324)	6676–8014 (1339)	−	ATA[a], ATG[b]	T*		11[a], −7[b]
nad4l	8007–8309 (303)	8008–8310 (303)	−	ATG	TAA		2
trnT	8312–8379 (68)	8313–8379 (67)	+			TGT	0
trnP	8380–8445 (66)	8380–8450 (71)	−			TGG	2
nad6	8448–8963 (516)	8453–8968 (516)	+	ATC[a], ATT[b]	TAA		−1
cob	8963–10,102 (1140)	8968–10,104 (1137)	+	ATG	TAG[a], TGA[b]		−5[a], −2[b]
trnS2	10,098–10164 (67)	10,103–10169 (67)	+			TGA	10[a], 31[b]
nad1	10,175–11128 (954)	10,201–11136 (936)	−	ATT	TAG[a], TAA[b]		27[a], 47[b]
trnL1	11,156–11,225 (70)	11,184–11,253 (70)	−			TAG	−44[a], −47[b]
rrnL (16S)	11,182–12,589 (1408)	11,213–12,603 (1391)	−				−7[a], 2[b]
trnV	12,583–12,653 (71)	12,606–12677 (72)	−			TAC	−3
rrnS (12S)	12,651–13,502 (852)	12,675–13,532 (858)	−				0
CR*	13,503–14235 (733)	13,533–14,256 (724)					0
trnI	14,236–14,302 (67)	14,257–14,322 (66)	+			GAT	−3
trnQ	14,300–14368 (69)	14,320–14388 (69)	−			TTG	8[a], 4[b]
trnM	14,377–14,445 (69)	14,393–14,462 (70)	+			CAT	30[a], 0[b]
nad2	14,476–15,447 (972)	14,463–15,464 (1002)	+	ATA[a], ATG[b]	TAA		−2[a], 0[b]
trnW	15,446–15,513 (68)	15,463–15,529 (67)	+			TCA	−1
trnC	15,513–15,578 (66)	15,529–15,593 (65)	−			GCA	0
trnY	15,579–15,644 (66)	15,594–15,659 (66)	−			TAC[a], GTA[b]	0

unusual start codon ACG. Twelve PCGs terminated with conventional stop codons (TAA or TGA) (see Table S2). The incomplete stop codon T was used for *nad4*. The PCGs in the mitochondrial genomes of both *P. echinatus* and *P. interruptus* were AT-rich and exhibited similar nucleotide usage. In *P. echinatus,* the A + T nucleotide composition was 60.8%. For *P. interruptus,* A + T nucleotide composition was 66.3%. Some variation in codon usage frequency for *P. echinatus* and *P. interruptus* was also revealed. In *P. echinatus* PCGs, the codons ATT (Ile, $n = 175$, 5.348%), TTC (Phe, $n = 169$, 5.165%), and TTT (Phe, $n = 139$, 4.248%) were among those used most frequently, while CGG (Arg, $n = 17$, 0.520%), AGC (Ser, $n = 13$, 0.397%), and CGC (Arg, $n = 2$, 0.061%) were used least frequently (excluding stop codons). Similarly, in *P. interruptus,* the codons ATT (Ile, $n = 241$, 6.470%), TTT (Phe, $n = 230$, 6.174%), and TTA

(Leu, $n = 226$, 6.067%) were used most frequently, while AGC (Ser, $n = 7$, 0.188%), CGC (Arg, $n = 3$, 0.081%), and CGG (Arg, $n = 3$, 0.081%) were least frequently observed (excluding stop codons).

The tRNA secondary structures of *P. echinatus* and *P. interruptus* (Fig. 2 & Appendix 1, respectively) were arranged in a cloverleaf formation – with the exception of Serine-1 – and ranged from 64 bp (trnL2) – 71 bp long (trnE & trnV) and 63 bp (trnD) – 72 bp long (trnV), respectively. In these two species, Serine-1 (*trnS-1*) had a reduced TΨC helical stem. Additionally, in *P. interruptus,* Serine-1 (*trnS1*) also possessed a TCT anticodon and Lysine (*trnK*) possessed a TTT anticodon.

The *rrnL* gene, located between *trnL1* and *trnV*, was 1,408 bp long in *P. echinatus* and 1,391 bp in *P. interruptus*. Meanwhile, the *rrnS* gene, located between *trnV* and the control region, was 852 bp long in *P. echinatus* and 858 bp in *P. interruptus*. These two genes were A + T rich. The nucleotide composition for *rrnL* was A = 34.9%, G = 12.3%, C = 20.0%, and T(U) = 32.8% in *P. echinatus* and A = 33.9%, G = 11.9%, C = 17.0%, and T(U) = 37.2% in *P. interruptus.* Likewise, the nucleotide composition for *rrnS* was A = 31.8%, G = 14.4%, C = 21.9%, and T(U) = 31.8% in *P. echinatus* and A = 33.3%, G = 12.2%, C = 18.5%, and T(U) = 35.9% in *P. interruptus*.

The control region (CR) nucleotide composition was A = 31.1%, G = 14.2%, C = 23.7%, and T(U) = 31.0% in *P. echinatus* and A = 35.9%, G = 11.6%, C = 16.6%, and T(U) = 35.9% in *P. interruptus*, indicating an A + T overrepresentation in the two species. Analysis of the

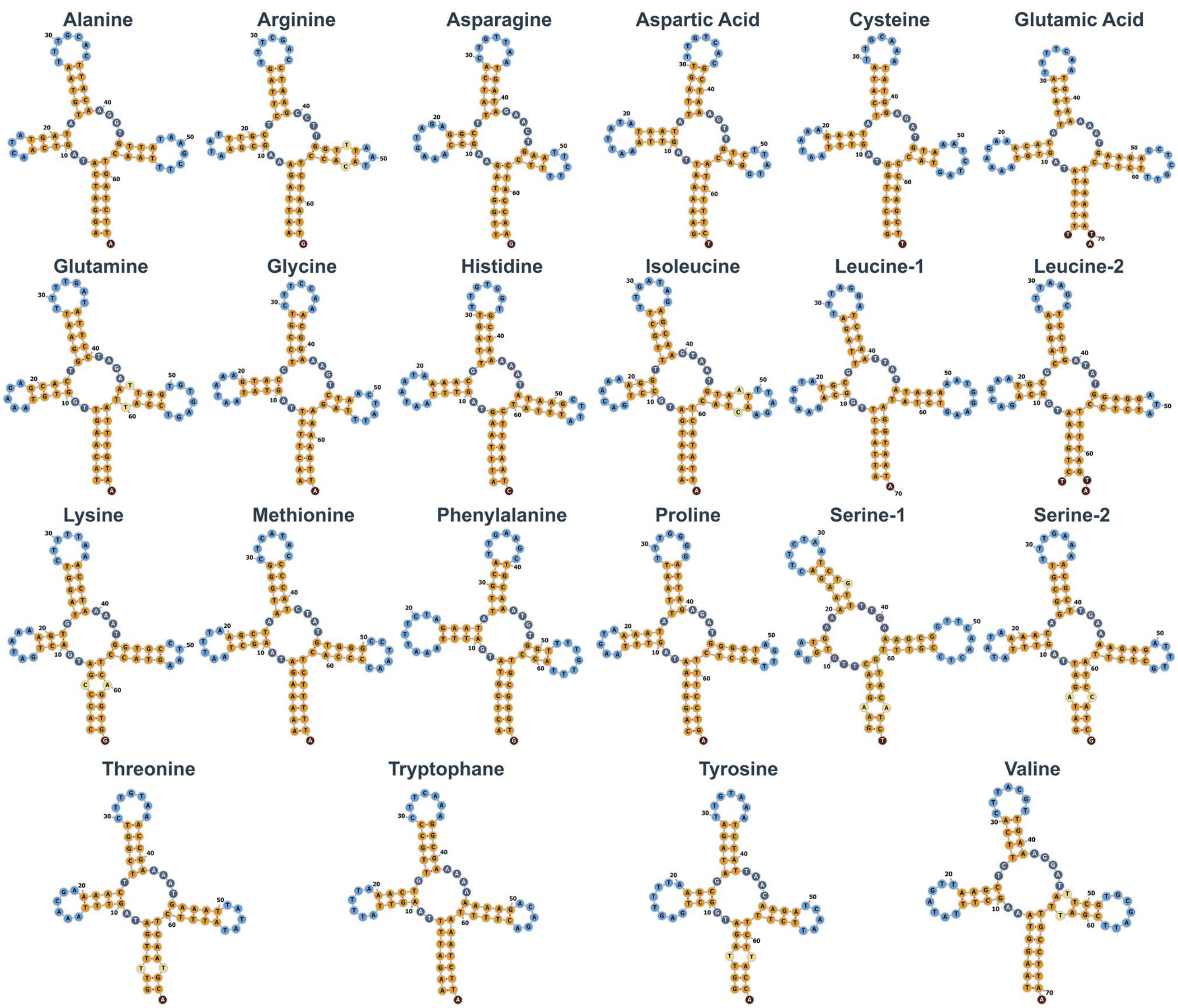

Fig. 2 Secondary tRNA showed a cloverleaf structure for *P. echinatus,* as estimated in MiTFi (Jühling et al. 2012) and depicted in the web server Forna (Kerpedjiev et al. 2015)

CR using Microsatellite Repeats Finder showed a total of 7 and 10 simple sequence repeats (SSRs) in *P. echinatus* and *P. interruptus*, respectively (see Table 2). In turn, no tandem repeats in the CR of *P. echinatus* and *P. interruptus* were identified by Tandem Repeats Finder. Lastly, the RNAstructure website generated 20 secondary structure predictions for the CR in the two studied species. The change in Gibbs free energy (ΔG) for these structures ranged from -88.8 kcal/mol to -88.0 kcal/mol for *P. echinatus* and -91.4 kcal/mol to -89.5 kcal/mol for *P. interruptus*. All secondary predictions exhibited variable numbers of stem-loops throughout the entire CR sequence.

Phylomitogenomics of the Achelata

The Maximum Likelihood phylogenetic analysis (see Fig. 3) confirmed the monophyly of the infraorder Achelata, considering that all members of the family Palinuridae – 16 species from the genera *Panulirus, Linuparus, Puerulus, Palinurellus,* and *Sagmariasus* – and all members of the Scyllaridae used in this study clustered into a fully supported clade (bv = 100). In the Achelata, the monophyletic status of the family Palinuridae was well supported (bv = 97) while the monophyly of the family Scyllaridae was fully supported. Within the family Scyllaridae, the subfamily Arctidinae (bv = 100), represented by three species of *Scyllarides* in our analysis, was sister to a clade containing representatives of the subfamilies Ibacinae (bv = 81) and Theninae (bv = 99).

Within the family Palinuridae, *Sagmariasus verreauxi* and *Phyllamphion elegans*, the only representatives of the 'Silentes' group with available mitogenomes in GenBank, occupied early branching positions. *Sagmariasus verreauxi* was positioned basal to a clade comprising all other Palinuridae, within which *P. elegans* was sister to another moderately supported clade (bv = 78) comprising all representatives of the family Palinuridae belonging to the 'Stridentes' group. The 'Stridentes' (bv = 75) comprised representatives from the genera *Linuparus, Puerulus,* and *Panulirus*. In the latter clade, *Linuparus trigonus* and *Puerulus angulatus* formed a fully supported clade that was sister to a fully monophyletic *Panulirus* (bv = 100).

In the family Palinuridae, the genus *Panulirus* consisted of the two fully supported (bv = 100) sister clades, Lineage 1 and Lineage 2 sensu George (2006). In Lineage 1, *Panulirus interruptus* had an early branching position and was arranged sister to a fully supported clade (bv = 100) containing the remaining species of Lineage 1 (*P. argus, P. echinatus, P. japonicus, P. longipes,* and *P. cygnus*). *Panulirus argus* was sister to a well-supported clade (bv = 98) containing *P. echinatus, P. japonicus, P. longipes* and *P. cygnus*. In Lineage 2, *P. versicolor* was basally positioned to a clade with poorly resolved relationships containing *P. polyphagus, P. stimpsoni, P. penicillatus,* and a fully supported clade (bv = 100) consisting of *P. homarus* and *P. ornatus*.

Table 2 Selection signatures across the 13 mitochondrial protein-coding genes in *Panulirus* spp. using Branch Models (BM), Branch-Site Models (BSM), and Site Models (SM) in the Datamonkey webserver. K denotes the selection intensity parameter; ω < 1 indicates negative (purifying) selection; ω > 1 indicates positive (diversifying) selection; > 0.95 indicates significant posterior probability

Gene	BM	BSM		SM		
	RELAX	aBSREL	BUSTED	MEME	FUBAR	SLAC
	K value (p-value; LR)	# branches (p-value)	# sites ER ≥ 10 (p-value)	sites ω > 1 (p-value)	# sites ω < 1 (> 0.95)	# sites ω < 1 (p < 0.05)
atp6	–	–	–	–	18	130
atp8	2.86 (0.000; 12.21)	–	–	–	25	16
cob	–	1 (0.042)	–	366 (0.02)	345	264
cox1	1.18 (0.002; 9.18)	3 (0.015)	–	505 (0.03); 510 (0.01); 512 (0.00)	475	366
cox2	1.73 (0.000; 36.35)	–	–	187 (0.04)	205	160
cox3	–	–	–	–	234	153
nad1	3.39 (0.000; 53.36)	1 (0.000)	2 (0.001)	–	275	195
nad2	2.09 (0.000; 17.23)	–	–	–	281	206
nad3	1.86 (0.001; 11.95)	1 (0.023)	–	–	101	81
nad4	–	1 (0.000)	–	10 (0.04)	381	245
nad4L	1.60 (0.038; 4.28)	–	–	–	80	49
nad5	1.43 (0.001; 11.06)	–	0 (0.046)	10 (0.04); 183 (0.02)	506	320
nad6	2.03 (0.000; 133.83)	1 (0.000)	–	–	141	95

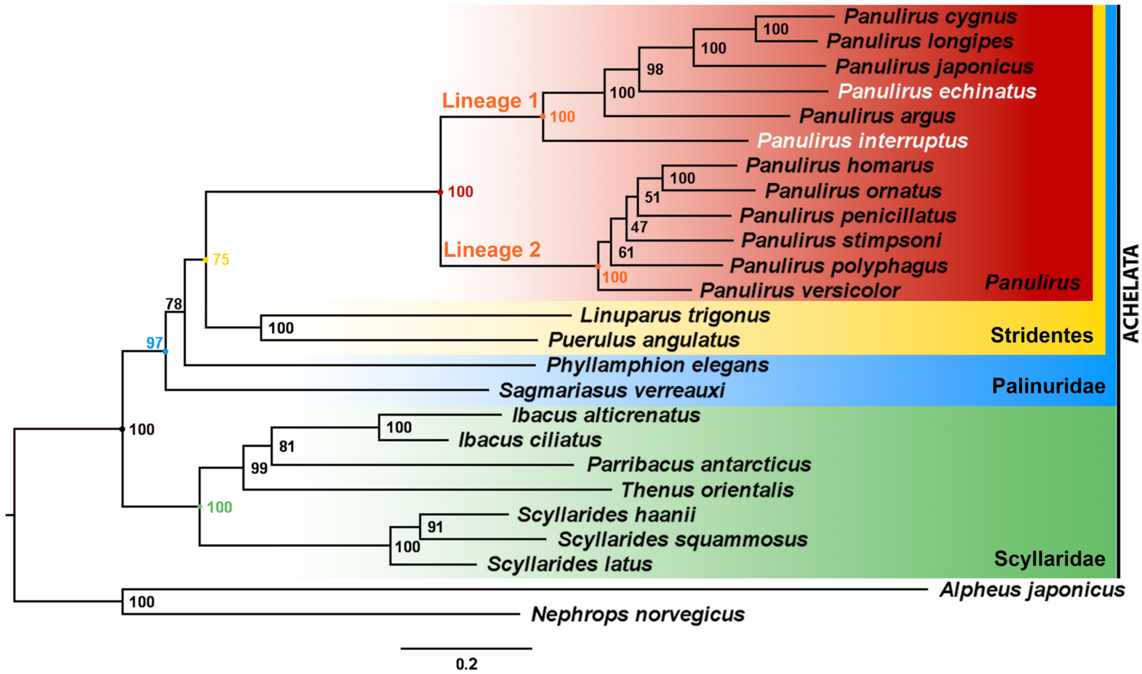

Fig. 3 Maximum likelihood phylogeny of the infraorder Achelata based on all mitochondrial protein-coding genes. We utilized only species with reported complete mitogenomes, namely 15 members of the family Palinuridae plus 7 species belonging to the family Scyllaridae. The outgroup included a member of the infraorder Caridea (*Alpheus japonicus*) and the infraorder Nephropidae (*Nephrops norvegicus*)

Adaptive evolution of mitochondrial protein-coding genes

In the K_A / K_S selective pressure analysis, the ω (K_A / K_S) ratio values for all mitochondrial PCGs of *P. echinatus* and *P. interruptus* were below 1 ($p < 0.001$), indicating evolution under purifying selection (see Fig. 3). In *P. echinatus,* the ω (K_A / K_S) ratios observed for *cox1, cox2,* and *cob* (ω = 0.00348, 0.00543, and 0.00448, respectively) were two orders of magnitude lower than the ω ratio calculated for *apt8* (ω = 0.10087), *nad6* (0.10266), and *nad2* (0.10276), which in turn, were higher than those observed for the remaining PCGs (min ω: 0.009615, max ω: 0.051561). This suggests stronger purifying selection and evolutionary constraints in the *cox1, cox2,* and *cob* genes and weaker purifying selection in the *apt8, nad6,* and *nad2* genes. In turn, in *P. interruptus,* the ω (K_A / K_S) ratios observed for *cob* (ω = 0.00817), *cox1* (0.00611), and *cox3* (0.00528) were two orders of magnitude lower than the ω ratio calculated for *atp8, nad5,* and *nad6* (ω = 0.05946, 0.04003, and 0.04002), which were higher than those observed for the remaining PCGs (min ω: 0.01229, max ω: 0.02776, respectively). This suggests strong purifying selection in the *cob, cox1,* and *cox3* genes and weaker purifying selection in the *atp8, nad6,* and *nad2* genes.

The RELAX analysis revealed that *atp8, cox1, cox2, nad1, nad2, nad3, nad4L, nad5,* and *nad6* experienced intensified purifying selection strength in tropical equatorial *Panulirus* spiny lobsters that reside in environments with relatively consistent temperatures (Table 2). In turn, *atp6* experienced relaxed selection strength in spiny lobsters belonging to the genus *Panulirus* (Table 2). Additionally, the RELAX analysis showed that *atp6, atp8, cob, nad2, nad3,* and *nad4L* experienced a relaxation in purifying selection strength (Table 3) in Scyllarid lobsters inhabiting low-oxygen depths. No evidence of selection strength either relaxing or intensifying along branches was found in *cox3* and *nad4* for either scenario (LRTs with $p > 0.05$).

The aBSREL analysis found evidence of episodic positive (diversifying) selection in 8 PCGs total. For equatorial *Panulirus*, *cob*, *cox1*, *nad1*, *nad3*, *nad4*, and *nad6* experienced episodic positive selection (Table 2). For *cob*, *nad3*, and *nad4*, the branch leading to the genus *Panulirus* was under episodic positive selection. *Cox1* also exhibited episodic positive selection on the branch leading to *Panulirus*, as well as on the branches leading to *Panulirus* Lineage 2 and *P. ornatus*. Additionally, *nad1* showed episodic positive selection on the branch leading to *P. homarus,* and *nad6* revealed episodic positive selection on the branch leading to *P. cygnus*. For Scyllaridae inhabiting low-oxygen depths, *atp6* and *nad5* experienced episodic positive selection (Table 3). In *atp6*, episodic positive selection occurred on the branch leading to the genus *Scyllarides* within the

Table 3 Selection signatures across the 13 mitochondrial protein-coding genes in species within the Scyllaridae using Branch Models (BM), Branch-Site Models (BSM), and Site Models (SM) in the Datamonkey webserver. K denotes the selection intensity parameter; ω < 1 indicates negative (purifying) selection; ω > 1 indicates positive (diversifying) selection; > 0.95 indicates significant posterior probability

Gene	BM	BSM		SM		
	RELAX	aBSREL	BUSTED	MEME	FUBAR	SLAC
	K value (p-value; LR)	# branches (p-value)	# sites ER ≥ 10 (p-value)	sites ω > 1 (p-value)	# sites ω < 1 (> 0.95)	# sites ω < 1 (p < 0.05)
atp6	0.75 (0.000; 16.78)	1 (0.007)	–	51 (0.04); 215 (0.04)	145	41
atp8	0.51 (0.002; 9.57)	–	–	–	10	4
cob	0.63 (0.000; 17.95)	–	–	18 (0.03); 120 (0.08)	289	108
cox1	–	–	–	512 (0.01)	436	141
cox2	–	–	–	131 (0.04); 227 (0.03)	167	64
cox3	–	–	–	–	202	74
nad1	–	–	–	42 (0.03)	225	82
nad2	0.68 (0.000; 14.15)	–	–	–	196	81
nad3	0.54 (0.000; 16.16)	–	–	–	74	27
nad4	–	–	–	–	294	109
nad4L	0.55 (0.000; 70.23)	–	–	–	54	15
nad5	–	1 (0.000)	–	33 (0.01); 379 (0.04); 547 (0.01)	372	148
nad6	–	–	–	–	151	120

Scyllaridae. In *nad5*, episodic positive selection occurred on the branch leading to Scyllaridae.

The BUSTED analysis also detected episodic positive (diversifying) selection in *nad1* ($p = 0.001$) and *nad5* ($p = 0.046$) for equatorial *Panulirus* experiencing consistent temperatures (Table 2). However, no evidence of episodic positive selection was found in any of the 13 PCGs for Scyllaridae inhabiting low-oxygen depths.

The MEME analysis revealed episodic positive selection on a variable number of sites depending upon the identity of the studied PCG. For equatorial *Panulirus* experiencing relatively constant temperatures, the number of sites under episodic positive selection ranged from 1 site in *cob, cox2, nad4,* and *nad6* to 3 sites in *cox1* (Table 2). For Scyllaridae inhabiting low-oxygen depths, the number of sites under episodic positive selection ranged from 1 site in *cox1* and *nad1* to 3 sites in *nad5* (Table 3).

The FUBAR analysis found no evidence of pervasive positive selection on any PCG. However, pervasive negative selection was detected in each PCG. In *Panulirus,* the number of sites under pervasive negative selection varied from 25 sites in *atp8* to 506 sites in *nad5* (Table 2). In Scyllaridae*,* the number of sites under pervasive negative selection varied from 10 sites in *atp8* to 436 sites in *cox1* (Table 3).

Lastly, the SLAC analysis found no evidence of pervasive positive selection on sites within protein-coding genes. Instead, pervasive negative selection was detected in all 13 PCGs, ranging from 16 sites in *atp8* to 366 sites in *cox1* for *Panulirus*, and from 4 sites in *atp8* to 148 in *nad5* for Scyllaridae (Tables 2 & 3, respectively).

Discussion

Mitochondrial Genomes of *Panulirus echinatus* and *Panulirus interruptus*

The mitochondrial genomes of *Panulirus echinatus* and *Panulirus interruptus* were AT-rich and compact with numerous overlapping genes and few gaps, in line with that reported for other decapod crustaceans, including other representatives of the Achelata (Baeza 2018 and references therein). The lengths of the two studied mitogenomes were also similar to those reported for other congeneric and cofamilial species. In the genus *Panulirus*, mitochondrial genome length varies between 15,665 bp in *P. homarus* (Xiao et al. 2017) and 15,739 bp in *P. argus* (Baeza 2018) while in the Achelata, mitochondrial genome length varies between 15,470 bp in *Sagmariasus verreauxi* (Doyle et al. 2015) and 16,826 bp in *Thenus orientalis* (Tan et al. 2016).

Mitochondrial gene arrangement in *P. echinatus* and *P. interruptus* (Fig. 1) was identical to that reported for the genus *Panulirus*, family Palinuridae, and infraorder Achelata (Liang 2012; Ahn et al. 2014; Yang et al. 2014; Doyle et al. 2015; Tan et al. 2016; Xiao et al. 2017; Baeza 2018; Andriyono et al. 2019). Additionally, start and stop codon usage in PCGs was similar to that reported before for other representatives of the Achelata. The use of ACG as a start codon by the *cox1* gene has been reported frequently in other species of *Panulirus* (Yamauchi et al. 2002; Liu & Cui 2011; Liang 2012; Shen et al. 2013; Kim et al. 2015; Xiao et al. 2017; Baeza 2018; Andriyono et al. 2019) and occasionally in other representatives of the infraorder Achelata (e.g. *Ibacus alticrenatus, Ibacus ciliatus, Scyllarides latus,* and *Thenus orientalis*) (NC_041153; Shen et al. 2013; Ahn et al. 2014; Tan et al. 2016). The incomplete stop codon T in *cox1* and *nad4* is also widespread in the genus *Panulirus* and family Palinuridae and, to a lesser extent, in the family Scyllaridae of the infraorder Achelata (Yamauchi et al. 2002; Liu & Cui 2011; Liang 2012; Ahn et al. 2014; Doyle et al. 2015; Yang et al. 2014; Kim et al. 2015; Baeza 2018; Andriyono et al. 2019; Liu et al. 2019; Yang et al. 2019b; Liu & Wang 2020; Liu et al. 2020; Liu & Pu 2021; Liu & Xia 2021). More importantly, protein-coding genes in *P. echinatus* and *P. interruptus* exhibited similar codon usage profiles; the most frequently used codons were A + T rich, while the least frequently used codons were C + G rich. Codons ending in thymine were favored over codons ending in adenine or cytosine, while codons ending in guanine were least used. Although codon usage has not been extensively evaluated in the Achelata, the aforementioned results agree with that reported for the two other species of *Panulirus* in which codon usage of mitochondrial PCGs has been studied (*P. argus*—Baeza 2018, *P. stimpsoni*—Liu & Cui 2011).

In the mitochondrial genomes of *P. echinatus* and *P. interruptus*, all but a single tRNA gene in the two studied species exhibited a 'cloverleaf'-shaped secondary structure, in line with that reported for *P. argus* (Baeza 2018), *P. japonicus* (Yamauchi et al. 2002), *P. ornatus* (Liang 2012), *P. polyphagus* (Andriyono et al. 2019), and *P. stimpsoni* (Liu & Cui 2011). Interestingly, truncation of tRNA-Serine-1 was observed in both *P. echinatus* and *P. interruptus*. Truncation of tRNA-Serine-1 commonly occurs in other eumetazoans (Jühling et al. 2012). Other *Panulirus* species, such as *P. argus, P. japonicus, P. ornatus*, and *P. stimpsoni* display a more drastic truncation of tRNA-Serine-1 given the complete elimination of the dynein heavy chain (DHC) (Yamauchi et al. 2002; Liu & Cui 2011; Liang 2012; Baeza 2018). The effect of truncation on crustacean tRNA-Serine-1 function is unknown, although it could potentially impact seryl-tRNA synthetase (SerRS), for which the tRNA-Serine-1 TΨC helical stem is the main recognition site (Ueda et al. 1992). tRNA-Serine-1 in *P. interruptus* also possessed a TCT anticodon, which is not a common occurrence in invertebrates (Li et al. 2016) but agrees with the Serine-1 anticodon observed in *P. japonicus* (Yamauchi et al. 2002) and *P. argus* (Baeza 2018) in the family Palinuridae, as well as in *Scyllarides squammosus* (Liu et al. 2019) and *S. haanii* (Liu & Wang 2020) in the family Scyllaridae.

The control regions in the two newly assembled mitochondrial genomes were A + T rich and contain microsatellites, but lack tandem repeats, in line with that reported for *P. argus* (Baeza 2018) (see Appendix 2). A + T rich CRs have also been observed in *P. homarus, P. japonicus, P. ornatus,* and *P. stimpsoni*. Except for *P. stimpsoni*, microsatellites and tandem repeats have not been evaluated in the aforementioned species (Yamauchi et al. 2002; Liu & Cui 2011; Liang 2012; Xiao et al. 2017). However, tandem repeats have been found in the CR of *P. stimpsoni* (Liu & Cui 2011). Microsatellites are also present in the control regions of *P. longipes, P. penicillatus, Linuparus trigonus, Parribacus antarcticus,* and *Scyllarides squammosus* (Liu et al. 2019; Yang et al. 2019b; Liu et al. 2020; Liu & Pu 2021; Liu & Xia 2021). Overall, information about the organization, or lack thereof, of the CR is limited in the Achelata and this region has been examined in detail by only three other studies focused on *P. stimpsoni* (Liu & Cui 2011), *P. japonicus* (Liu & Cui 2011), and *P. argus* (Baeza 2018). Generally, control region organization varies within the *Panulirus* genus. In *P. stimpsoni*, the CR contains one [TA(A)]n-block, two GA-blocks and three hairpin structures (Liu & Cui 2011). The CR in *P. japonicus* is similar, with one [TA(A)]n-block, two GA-blocks and two hairpin structures, albeit in different positions than for *P. stimpsoni* (Liu & Cui 2011). In *P. argus*, the CR has no obvious organization, with only two conserved motifs (Baeza 2018). Additional studies on CRs belonging to other spiny lobsters and decapod crustaceans are necessary to enhance the understanding of the role of this non-coding region in the translation and replication of mitochondrial genomes.

Phylomitogenomics of the Achelata

Our ML phylogenetic tree (Fig. 3) recovered a monophyletic Achelata, in line to that reported before by Ptacek et al. (2001), Patek & Oakley (2003), Palero et al. (2009), Bracken-Grissom et al. (2014), and Davis et al. (2015). The aforementioned studies used either only short mitochondrial fragments or a combination of mitochondrial and nuclear markers for phylogenetic inference. However, there were three major differences among our ML phylogeny based on mitochondrial PCGs and previous studies (Ptacek et al. 2001; Patek & Oakley 2003; Palero et al. 2009; Bracken-Grissom et al. 2014; Davis et al. 2015; Palero et al. 2016). First, in the family Scyllaridae and within the clade that contained the genera *Thenus + Parribacus + Ibacus*, *Thenus orientalis* exhibited an early branching position, while in the phylogenetic tree recovered by Bracken-Grissom et al. (2014), Davis et al. (2015), and Palero et al. (2016), *Ibacus* held an early branching position in the same clade.

Second, in the family Palinuridae, *Sagmariasus verreauxi* and *Phyllamphion elegans* were not part of a well-supported 'Silentes' reported by Palero et al. (2009) and Bracken-Grissom et al. (2014). Thus, our results do not support the split between 'Silentes' and 'Stridentes' in spiny lobsters (Palero et al. 2009; Bracken-Grissom et al. 2014; Davis et al. 2015). In our ML tree, *Sagmariasus verreauxi* and *Phyllamphion elegans* held early branching positions in the family Palinuridae. Of these two species, *S. verreauxi* was placed most basal in the Palinuridae. In contrast to our results, phylogenetic relationships within the Palinuridae retrieved by Palero et al. (2009 and Bracken-Grissom et al. (2014) supported the split between 'Silentes' and 'Stridentes' in spiny lobsters with *S. verreauxi* and *P. elegans* forming part of a well-supported Silentes clade. Additionally, *P. elegans* was placed more basal in the 'Silentes' clade than *S. verreauxi*. These disagreements might be due to the lack of complete mitogenomes for most 'Silentes' group species in our analysis. Indeed, the position of *P. elegans* in our phylogeny was only moderately supported.

Third, within the genus *Panulirus*, our phylogeny placed *P. penicillatus* within Lineage 2. However, all prior phylogenies of *Panulirus* have retrieved *P. penicillatus* within Lineage 1 (Ptacek et al. 2001; Patek & Oakley 2003; Palero et al. 2009; Bracken-Grissom et al. 2014; Davis et al. 2015). The difference between our tree and those previously reported by the authors mentioned above likely stems from either our study missing data for numerous species previously documented in Lineage 2 or was due to specimen misidentification (in the case of *P. penicillatus* sequenced by Liu & Xia (2021)). Most of the nodes within Lineage 2 of our phylogeny – including the node representing the most recent common ancestor of *P. penicillatus* – were poorly resolved. Overall, more mitogenomes (and nuclear markers; e.g., ribosomal genes) are needed to improve our understanding of phylogenetic relationships in the Achelata and resolve ambiguous relationships within the family Palinuridae and genus *Panulirus*.

Pervasive purifying selection in mitochondrial protein-coding genes in the Achelata

In the mitochondrial genomes of *P. echinatus* and *P. interruptus*, all protein-coding genes were exposed to purifying selection as indicated by our first K_A / K_S selective pressure analysis, as well as additional FUBAR and SLAC analyses conducted in Datamonkey for all representatives of the Achelata. Selective pressures on mitochondrial PCGs are poorly studied in the Achelata and other decapod crustaceans with sequenced mitochondrial genomes (Baeza 2018). Nonetheless, our results agree with the robust signature of purifying selection detected for mitochondrial PCGs in the spiny lobster *Panulirus argus* (Baeza 2018), the only other species of spiny lobsters in which an analysis of selective pressure has been conducted before. Our results are also in line with that reported for a few other decapod crustaceans in which selective pressures in PCGs have been examined (i.e. caridean shrimp: *Synalpheus microneptunus*—Chak et al. 2020, *Pandalus platyceros*—Cronin et al. 2022b; penaeid shrimps: *Xiphopenaeus kroyeri*—Cronin et al. 2022a, *Fenneropenaeus penicillatus*—Zhang et al. 2015, and *Farfantepenaeus duorarum*—Collins et al. 2022; brachyuran crabs: *Metopaulias depressus* —Rodriguez-Pilco et al. 2022). Overall, our results support the notion that purifying selection is pervasive in mitochondrial PCGs and likely curbs the potential for adaptive evolution of PCGs in order to preserve the functionality of the OXPHOS pathway, which is crucial for energy production via cellular respiration (Bernt et al. 2013).

Signatures of adaptive evolution in protein-coding genes of spiny lobster *Panulirus* spp.

In *Panulirus* spp., all 13 PCGs contained sites under pervasive negative / purifying selection (per SMs), which suggests that the evolution of PCGs is primarily constrained to preserve functionality of the OXPHOS pathway. Over the strong negative selection background detected for all genes in this study, spiny lobsters *Panulirus* spp. inhabiting equatorial latitudes with consistent high temperatures were predicted to exhibit signatures of adaptive evolution on mitochondrial protein-coding genes. Specifically, we predicted positive selection in the Complex I subunits *nad2*, *nad4*, and *nad6*, considering that Complex I functionality is temperature-sensitive, and climate-specificity likely drives the optimization of this proton pump for specific local temperature fluctuations (Stillman 2002; Iftikar et al. 2014; Oellermann et al. 2020). Our evaluation of positive selection in mitochondrial PCGs in the Achelata with BMs and BSMs supported this prediction. In *Panulirus* spp. signatures of positive selection were found in all Complex I subunits. BMs indicated that purifying selection intensified for each subunit except *nad4*, and BSMs confirmed episodic positive selection in *nad1, nad3, nad4, nad5,* and *nad6*. Furthermore, in Complex I, *nad4* and *nad5* contained sites under episodic positive selection as detected by one of our SM analyses. These findings indicate that Complex I genes are likely experiencing adaptive evolution due to environmental pressures (i.e. high and relatively steady temperatures) characteristic of high temperature tropical environments. Our results therefore suggest that adaptive evolution in tropical *Panulirus* spiny lobsters likely preserves Complex I functionality within the narrow temperature ranges of equatorial climates.

Similar signatures of positive selection in Complex I have been reported for other marine crustaceans that live in environments with high (and stable) temperatures. For instance, studies on deep sea hydrothermal vent crustaceans (depths > 1500 m) suggest that Complex I genes experience adaptive evolution due to extremely stable elevated temperatures year-round (up to 390 ^{o}C) (Wang et al. 2017; Sun et al. 2018, 2019a, b, 2021). However, hydrothermal vents are also characterized by hypoxia and elevated levels of hydrogen sulfide, methane, and heavy metals (Little & Vrijenhoek 2003). Thus, adaptive pressures from the aforementioned conditions might also explain positive selection on Complex I genes in hydrothermal vent crustaceans.

Furthermore, positive selection in Complex I is not exclusively linked to elevated temperatures. Extensive positive selection on Complex I genes has been observed in other invertebrates that inhabit environments with consistently low temperatures. For example, research on mayflies with an extensive coldwater aquatic larval phase (< 0 to 10 ^{o}C) indicated that mitochondrial Complex I PCGs also experience extensive positive selection (Xu et al. 2021). This suggests that although positive selection in Complex I is influenced by temperature, adaptation in Complex I PCGs is not exclusive to hot or cold environments. Instead, it is likely that Complex I PCGs adapt due to stress from prolonged exposure to narrow temperature ranges.

Interestingly, besides that reported for Complex I, signatures of positive selection were also found in Complex III, IV, and V mitochondrial PCGs of *Panulirus* spp. inhabiting equatorial latitudes with consistently high temperatures. Similar signatures of positive selection in Complex III, IV, and V have previously been found in other marine crustaceans – including shrimps, crabs, and squat lobsters – that experience narrow temperature ranges. Various studies suggest hydrothermal vent crustaceans experience adaptive evolution on Complex III (*cob*), Complex IV (including *cox1* and *cox2*) and Complex V (including *atp8*) in response to prolonged extreme temperatures (up to 390 ^{o}C) (Wang et al. 2017; Sun et al. 2018, 2019a, b, 2021). Adaptive evolution is likewise reported in the mitochondrial PCGs of other invertebrates that endure relatively constant temperatures. For instance, hydrothermal vent bivalves also exhibit positive selection on Complex III and Complex IV (Yang et al. 2019a, b). Similarly, coldwater mayflies undergo adaptive evolution on Complex III and Complex IV genes in response to constant exposure to cold temperatures (< 0 to 10 ^{o}C) (Xu et al. 2021). Adaptive evolution likely retains mutations that improve protein stability at sustained temperatures, thus enhancing the ability of Complex III to conduct electron transfer and proton translocation (DePristo et al. 2005; Bromberg & Rost 2009; Luo et al. 2013; Chaban et al. 2014). Similar mutations affecting protein stability in Complex IV might additionally impact the efficiency that Complex IV transfers electrons to oxygen to form water (Luo et al. 2013; Chaban et al. 2014). It is also likely that Complex V PCG adaptations influence overall energy production, as Complex V is the final protein in the OXPHOS pathway and is responsible for energy production via oxidative phosphorylation (Lehninger 1964; Luo et al. 2013; Chaban et al. 2014; Wang et al. 2017; Sun et al. 2018, 2019a, b, 2021). In summary, these studies demonstrate that in order to preserve normal cellular activities in environments with narrow temperature ranges, positive selection on mitochondrial PCGs likely optimizes the functionality of all OXPHOS pathway complexes.

Signatures of adaptive evolution in protein-coding genes of slipper lobsters

In the family Scyllaridae, pervasive negative / purifying selection was also found in all 13 PCGs (according to SMs), suggesting that the evolution of PCGs is primarily constrained to preserve the functionality of the OXPHOS pathway. Slipper lobsters (family Scyllaridae), which inhabit depths characterized by low-oxygen levels, were also predicted to experience adaptive evolution on mitochondrial protein-coding genes over pervasive negative selection background detected for all PCGs. Specifically, we expected positive selection in Complex V genes, namely the *atp6* and *atp8* subunits, considering that Complex V experiences increased energy production demands in perpetually low-oxygen conditions such as the relatively deep habitats inhabited by slipper lobsters, and thus mutations likely optimize enzyme efficiency to maintain cellular function under these low-oxygen conditions. Our tests for positive selection in PCGs using BM and BSM models in the Achelata supported this prediction. Within the family Scyllaridae, signatures of positive selection were observed in both Complex V mitochondrial-encoded subunits. BMs showed that purifying selection relaxed in *atp6* and *atp8*, and BSMs confirmed episodic positive selection in *atp6*. Moreover, one of our SMs showed that *atp6* in Complex V contained sites potentially under episodic positive selection. These tests indicate that Complex V genes likely undergo adaptive evolution due to environmental pressures on species inhabiting low-oxygen depths. Our results suggest that in deepwater slipper lobsters, adaptive evolution likely refines the ability of Complex V to produce energy in low-oxygen conditions.

Previous studies have reported similar signatures of positive selection in Complex V for other marine invertebrates inhabiting low-oxygen environments. Hypoxia has been proposed to drive adaptive evolution in Complex V genes in cave shrimps (Guo et al. 2018), deep sea black corals (depths < 200 m) (Ramos et al. 2023), and deep sea crustaceans from hydrothermal vent communities (Wang et al. 2017; Sun et al. 2018, 2019a, b, 2021). However, as mentioned previously, adaptive pressures from high temperatures and elevated levels of hydrogen sulfide, methane, and heavy metals might also explain positive selection on Complex V in deep sea hydrothermal vent crustaceans (Little & Vrijenhoek 2003). Likewise, deep sea black corals also experience high hydrostatic pressure and cold temperatures (Danovaro et al. 2014). As such, these conditions offer other possible explanations for positive selection on Complex V in deep sea black corals.

Positive selection in Complex V due to hypoxia is not merely limited to crustaceans nor the deep sea. Extensive positive selection on Complex V genes has also been described in a variety of other groups experiencing low-oxygen conditions. For instance, endosymbiotic and parasitic flatworms experience relaxed purifying selection on Complex V PCGs that might be explained by – but is not yet confirmed as – a response to low-oxygen levels in the digestive system of their hosts (Monnens et al. 2020). There are also numerous examples of positive selection on Complex V in high altitude species (1300 to 4800 m above sea level) subjected to hypoxic conditions, including brine shrimps (*Artemia)* (Zhang et al. 2013), ground birds (Zhou et al. 2014), and Tibetan horses (Xu et al. 2007). The diversity of species experiencing similar signatures of positive selection on Complex V suggests that adaptation occurs in Complex V PCGs due to a common stressor. Our results therefore suggest that Complex V PCGs likely adapt due to low oxygen or hypoxic conditions inherent to the inhabited environment across animals with disparate evolutionary origins.

In addition to Complex V, positive selection was also found in Complex I, III, and IV mitochondrial PCGs in Scyllaridae inhabiting depths with low-oxygen levels. Similar patterns of positive selection in low-oxygen conditions are also found in other marine invertebrates. For instance, previous studies indicate that hydrothermal vent and cave crustaceans (Wang et al. 2017; Guo et al. 2018; Sun et al. 2018, 2019a, b, 2021) and deep sea corals (Ramos et al. 2023) experience adaptive evolution on Complex I (including *nad1* and *nad5*), Complex III (*cob*), and Complex IV (including *cox1* and *cox2*) in response to hypoxia at depth. Similarly, fishes (Li et al. 2013) and mammals Xu et al. 2007; Xu et al. 2007; Luo et al. 2013; Yu et al. 2011; Luo et al. 2013) inhabiting high altitude habitats also undergo adaptive evolution on mitochondrial genes belonging to Complex I, Complex III, and Complex IV due to hypobaric hypoxia. Adaptive evolution likely enhances mitochondrial capacity for energy metabolism in low-oxygen conditions by favoring mutations that improve the ability of Complex I to pump protons into the mitochondrial intermembrane space (Wang et al. 2017; Guo et al. 2018). Likewise, adaptive evolution under hypoxic conditions likely preserves mutations that improve electron transfer and proton translocation in Complex III and boosts the rate that Complex IV transfers electrons to oxygen to form water (DePristo et al. 2005; Bromberg & Rost 2009; Luo et al. 2013; Chaban et al. 2014). Overall, our results and prior research indicate that mitochondrial PCGs in all OXPHOS pathway complexes undergo positive selection in low-oxygen environments.

Despite our evidence for positive selection on mitochondrial protein-coding gene complexes affected by narrow temperature ranges and hypoxia, other environmental factors could also explain these adaptive evolution patterns in spiny and slipper lobsters. For instance, shallow tropical equatorial waters inhabited by *Panulirus* spp. are also characterized by elevated levels of food availability, dissolved oxygen (200 µmol/kg – 260 µmol/kg), turbidity, and sunlight alongside variable salinity (0 ppt – 37 ppt) (Hatcher et al. 1989; Garcia et al. 2018; Stammer et al. 2021). Shallow habitats are also easily impacted by runoff and human activity (Hatcher et al. 1989). Meanwhile, the deep sea waters inhabited by Scyllaridae exhibit limited food availability, cold temperatures (0 °C – 3 °C, excluding hydrothermal vents habitats), high pressure (83.27 atm at 850 m), stable salinity (35 ppt), and a lack of sunlight (Ohmae et al. 2013; Danovaro et al. 2014; Stammer et al. 2021). Many species within Scyllaridae also have ranges that extend into both shallow and deep regions and are thus potentially exposed to a wider variety of selective pressures than those evaluated herein (Holthuis 1991; Lavalli et al. 2019). Future research on adaptive evolution within the Achelata should consider expanding available mitogenome data to evaluate these environmental pressures on subsets within spiny and slipper lobsters in more detail.

Selection analysis limitations and further considerations

Although our study evaluated adaptive evolution in mitochondrial protein-coding genes based on selection on nucleotide substitutions, adaptive evolution can also occur via gene deletion (loss of nucleotides), duplication (multiple copies of a particular region or gene), or differential expression (Olson-Manning et al. 2012; Yanchukov & Proulx 2012). Our study did not detect gene deletions or duplications among the species analyzed, but mechanisms modulating the gene expression of mitochondrial genes remain to be addressed. To further clarify how mitochondrial PCGs adapt to narrow temperature gradients and low-oxygen environments in the Achelata, crustaceans, and beyond, we suggest that future studies examine mitochondrial PCG expression in detail.

Our study explored selective pressures at the species level. Importantly, however, different populations from the same species might also experience disparate selective pressures or respond differently to the same pressures (Stanley 1975; Hohenlohe et al. 2010a, b). Patterns of genetic variation that are not well explained at the species level might be resolved by population level selective pressure analyses, which provide insights into genomic differences within and among populations (Wakeley 2009). Furthermore, although selective pressure analyses at the species level are useful in identifying loci under selection, as we have done here, we argue that subsequent functional studies are needed to clarify the relationship between specific single nucleotide variants in particular PCGs and environmental conditions, namely such as temperature and hypoxia, studied herein (Sabeti et al. 2006; Hassanin et al. 2009; Hohenlohe et al. 2010b). As such, we recommend that future studies aiming to improve our understanding of how species adapt to narrow temperature ranges and hypoxia also consider conducting selective pressure analyses at the population level.

In conclusion, our study assembled the complete mitogenomes of the brown spiny lobster, *P. echinatus*, and the California spiny lobster, *P. interruptus*. These species are the targets of major fisheries and new genomic resources will aid in conservation and management efforts. Next, we investigated whether adaptive evolution occurred in the 13 mitochondrial PCGs of spiny lobsters in the genus *Panulirus* inhabiting tropical equatorial regions with relatively consistent high temperatures and of representatives of the slipper lobster family Scyllaridae inhabiting low-oxygen deep environments. We detected strong background purifying selection across all mitochondrial PCGs in *P. echinatus* and *P. interruptus*, as well as other *Panulirus* species inhabiting equatorial latitudes and Scyllaridae inhabiting low-oxygen depths. Over the strong negative selection background observed for all PCGs in this study, signatures of positive selective pressure were detected in mitochondrial PCGs in equatorial *Panulirus* spp. and deepwater Scyllaridae. Similar signatures of adaptive evolution are also observed in other groups that endure narrow temperature ranges or hypoxia, such as coldwater mayflies (Xu et al. 2021), endoparasitic flatworms (Monnens et al. 2020), and high altitude artemia (Zhang et al. 2013), fishes (Li et al. 2013), ground birds (Zhou et al. 2014), and horses (Xu et al. 2007). Overall, our results are in line with other studies evaluating and finding adaptive evolution in mitochondrial PCGs in species inhabiting caves (shrimps—Guo et al. 2018), deep sea hydrothermal vents (shrimps, crabs, and squat lobsters—Wang et al. 2017; Sun et al. 2018, 2019a, b, 2021), and deep sea corals (Ramos et al. 2023). Thus, our study supports the notion that mitochondrial protein-coding genes can and do experience adaptive evolution to optimize mitochondrial function in a wide variety of organisms. Further research is needed to clarify which and how environmental conditions influence the adaptive evolution of mitochondrial PCGs in other crustacean lineages and beyond.

Acknowledgements To Dr. Miguel Angel del Río Portilla from CICESE Mexico for providing the sample of *Panulirus interruptus*. Funding was received from CIBNOR and CICESE through the SUBNARGENA and SAGARPA of Mexico and CONACYT grand 2005-12058 to FJGdL. This research was conducted as part of AMB's MSc thesis at Clemson University.

Funding Open access funding provided by the Carolinas Consortium.

References

Abascal F, Zardoya R, Posada D (2005) ProtTest: selection of best-fit models of protein evolution. Bioinformatics 21(9):2104–2105. https://doi.org/10.1093/bioinformatics/bti263

Abele D, Philipp E, Gonzalez PM, Puntarulo S (2007) Marine invertebrate mitochondria and oxidative stress. Front Biosci 12:933–946

Ahn D, Kim S, Park J, Shin S, Min G (2014) The complete mitochondrial genome of the Japanese fan lobster *Ibacus ciliatus* (Crustacea, Achelata, Scyllaridae). Mitochondrial DNA Part A 27(3):1871–1873. https://doi.org/10.3109/19401736.2014.971265

Andriyono S, Alam M, Pramono H, Abdillah AA, Kim HW (2019) Next-generation sequencing yields the complete mitochondrial genome of mud spiny lobster *Panulirus polyphagus* (Crustacea: Decapoda) from Madura water. Earth Environ Sci https://doi.org/10.1088/1755-1315/348/1/012020

Artimo P, Jonnalagedda M, Arnold K, Baratin D, Csardi G, de Castro E, Duvaud S, Flegel V, Fortier A, Gasteiger E, Grosdidier A, Hernandez C, Ioannidis V, Kuznetsov D, Liechti R, Moretti S, Mostaguir K, Redaschi N, Rossier G, Xenarios I, Stockinger H (2012) ExPASy: SIB bioinformatics resource portal. Nucleic Acids Res 40(W1):W597–W603. https://doi.org/10.1093/nar/gks400

Baeza A (2018) The complete mitochondrial genome of the caribbean spiny lobster *Panulirus argus*. Nat Sci Rep 8:17690. https://doi.org/10.1038/s41598-018-36132-6

Benson G (1999) Tandem repeats finder: a program to analyze DNA sequences. Nucleic Acids Res 27(2):573–580

Bernt M, Donath A, Jühling F, Externbrink F, Florentz C, Fritzsch G, Pütz J, Middendorf M, Stadler PF (2013) MITOS: improved de novo metazoan mitochondrial genome annotation. Mol Phylogenet Evol 69(2):313–319

Bracken-Grissom HD, Ahyong ST, Wilkinson RD, Feldmann RM, Schweitzer CE, Breinholt JW, Bendall M, Palero F, Chan T-Y, Felder DL, Robles R, Chu K-H, Tsang L-M, Kim D, Martin JW, Crandall KA (2014) The emergence of lobsters: phylogenetic relationships, morphological evolution and divergence time comparisons of an ancient group (Decapoda: Achelata, Astacidea, Glypheidea, Polychelida). Syst Biol 63(4):457–479. https://doi.org/10.1093/sysbio/syu008

Bromberg Y, Rost B (2009) Correlating protein function and stability through the analysis of single amino acid substitutions. BMC Bioinform 10(8):S8. https://doi.org/10.1186/1471-2105-10-S8-S8

Capella-Gutiérrez S, Silla-Martínez JM, Gabaldón T (2009) trimAl: a tool for automated alignment trimming in large-scale phylogenetic analyses. Bioinformatics 25(15):1972–1973. https://doi.org/10.1093/bioinformatics/btp348

Castro, P, Davie P, Guinot D, Schram F, von Vaupel Klein C (eds) (2015) Treatise on Zoology-Anatomy, Taxonomy, Biology. The Crustacea, Volume 9 Part C (2 Vols): Brachyura. Brill

Cecchini G (2003) Function and structure of complex II of the respiratory chain. Annu Rev Biochem 72:77–109. https://doi.org/10.1146/annurev.biochem.72.121801.161700

Chaban Y, Boekema EJ, Dudkina NV (2014) Structures of mitochondrial oxidative phosphorylation supercomplexes and mechanisms for their stabilisation. Biochem Biophys Acta 1837(4):418–426. https://doi.org/10.1016/j.bbabio.2013.10.004

Chak STC, Barden P, Baeza JA (2020) The complete mitochondrial genome of the eusocial sponge-dwelling snapping shrimp Synalpheus microneptunus. Sci Rep 10:7744. https://doi.org/10.1038/s41598-020-64269-w

Chan T-Y (2010) Annotated checklist of the world's marine lobsters (Crustacea: Decapoda: Astacidea,Glypheidea, Achelata, Polychelida). Raffles Bull Zool 23:153–181

Christen F, Desrosiers V, Dupont-Cyr BA, Vandenberg GW, Le François NR, Tardif J-C, Dufresne F, Lamarree SG, Blier PU (2018) Thermal tolerance and thermal sensitivity of heart mitochondria: mitochondrial integrity and ROS production. Free Radical Biol Med 116:11–18. https://doi.org/10.1016/j.freeradbiomed.2017.12.037

Collins SB, Bracken-Grissom HD, Baeza JA (2022) The complete mitochondrial genome of the pink shrimp Farfantepenaeus duorarum (Burkenroad 1939) (Decapoda: Dendrobranchiata: Penaeidae). J Crustac Biol 42(1):ruac007. https://doi.org/10.1093/jcbiol/ruac007

Conant GC, Wolfe KH (2008) GenomeVx: simple web-based creation of editable circular chromosome maps. Bioinformatics 24(6):861–862. https://doi.org/10.1093/bioinformatics/btm598

Cronin TJ, Conrad I, Kerkhove TRH, Hellemans B, De Troch M, Volckaert FAM, Baeza JA (2022b) Characterization of the complete mitochondrial genome of the Atlantic seabob shrimp Xiphopenaeus kroyeri Heller, 1862 (Decapoda: Dendrobranchiata: Penaeidae), with insights into the phylogeny of Penaeidae. J Crustacean Biol 42(1):ruac004. https://doi.org/10.1093/jcbiol/ruac004

Cronin TJ, Jones SJM, Baeza JA (2022b) The complete mitochondrial genome of the spot prawn, Pandalus platyceros Brandt

Cucini C, Leo C, Iannotti N, Boschi S, Brunetti C, Pons J, Fanciulli PP, Frati F, Carapelli A, Nardi F (2021) EZmito: a simple and fast tool for multiple mitogenome analyses. Mitochondrial DNA Part B 6(3):1101–1109. https://doi.org/10.1080/23802359.2021.1899865

Danovaro R, Snelgrove PVR, Tyler P (2014) Challenging the paradigms of deep-sea ecology. Trends Ecol Evol 29(8):465–475. https://doi.org/10.1016/j.tree.2014.06.002

Darriba D, Taboada LG, Doallo R, Posada D (2012) jModelTest 2: more models, new heuristics and high-performance computing. Nat Methods 9(8):772. https://doi.org/10.1038/nmeth.2109

Davis KE, Hesketh TW, Delmer C, Wills MA (2015) Towards a supertree of arthropoda: a species-level supertree of the spiny, slipper and coral lobsters (decapoda: achelata). PLoS ONE 10(10):e0140110. https://doi.org/10.1371/journal.pone.0140110

DePristo MA, Weinreich DM, Hartl DL (2005) Missense meanderings in sequence space: a biophysical view of protein evolution. Nat Rev Genet 6:678–687. https://doi.org/10.1038/nrg1672

Delport W, Poon AFY, Frost SDW, Pond SLK (2010) Datamonkey 2010: a suite of phylogenetic analysis tools for evolutionary biology. Bioinform Appl Note 26(19):2455–2457. https://doi.org/10.1093/bioinformatics/btq429

Devin A, Rigoulet M (2007) Mechanisms of mitochondrial response to variations in energy demand in eukaryotic cells. Am J Physiol Cell Physiol 292:C52–C58. https://doi.org/10.1152/ajpcell.00208.2006

Donath A, Jühling F, Al-Arab M, Bernhart SH, Reinhardt F, Stadler PF, Middendorf M, Bernt M (2019) Improved annotation of protein-coding genes boundaries in metazoan mitochondrial genomes. Nucleic Acids Res 47(20):10543–10552

Doyle S, Griffith I, Murphy N, Strugnell J (2015) Low-coverage MiSeq next generation sequencing reveals the mitochondrial genome of the Eastern rock lobster. *Sagmariasus verreauxi* Mitochondrial DNA 26(6):844–845. https://doi.org/10.3109/19401736.2013.855921

Dunn CD (2020) Sequencebouncer: a method to remove outlier entries from a multiple sequence alignment. BioRxiv https://doi.org/10.1101/2020.11.24.395459

Edgar RC (2004a) Muscle: a multiple sequence alignment method with reduced time and space complexity. BMC Bioinformatics 5:113. https://doi.org/10.1186/1471-2105-5-113

Edgar RC (2004b) MUSCLE: multiple sequence alignment with high accuracy and high throughput. Nucleic Acids Res 32(5):1792–1797. https://doi.org/10.1093/nar/gkh340

Ekström A, Sandblom E, Blier PU, Cyr B-AD, Brijs J, Pichaud N (2017) Thermal sensitivity and phenotypic plasticity of cardiac mitochondrial metabolism in European perch, *Perca fluviatilis*. J Exp Biol 220:386–396. https://doi.org/10.1242/jeb.150698

Fackler C (2019) Spiny lobster (Panulirus interruptus) at Channel Islands NMS, California. NOAA National Marine Sanctuaries. [Photograph]

Farrell AP (2002) Cardiorespiratory performance in salmonids during exercise at high temperature: insights into cardiovascular design limitations in fishes. Comp Biochem Physiol 132A:797–810

da Fonseca RRD, Johnson WE, O'Brien SJ, Ramos MJ, Antunes A (2008) The adaptive evolution of the mammalian mitochondrial genome. BMC Genom 9:119. https://doi.org/10.1186/1471-2164-9-119

Gaeta J (2013) *Panulirus echinatus* Smith, 1869 at Farol pool, Rocas Atoll, 2013. [Photograph]

Garcia HE, Weathers K, Paver CR, Smolyar I, Boyer TP, Locarnini RA, Zweng MM, Mishonov AV, Baranova OK, Seidov D, Reagan JR (2018) World Ocean Atlas 2018, Volume 3: Dissolved Oxygen, Apparent Oxygen Utilization, and Oxygen Saturation. A. Mishonov Technical Ed.; NOAA Atlas NESDIS 83, 38pp

George RW, Main AR (1967) The evolution of spiny lobsters (palinuridae): a study of evolution in the marine environment. Evolution 21(4):803–820. https://doi.org/10.1111/j.1558-5646.1967.tb03435.x

George RW (2006) Tethys Origin and Subsequent Radiation of the Spiny Lobsters (Palinuridae). Crustaceana 79(4):397–422. https://www.jstor.org/stable/20107668

Guo H, Yang H, Tao Y, Tang D, Wu Q, Wang Z, Tang B (2018) Mitochondrial OXPHOS genes provides insights into genetics basis of hypoxia adaptation in anchialine cave shrimps. Genet Genom 40:1169–1180. https://doi.org/10.1007/s13258-018-0674-4

Hassanin A, Ropiquet A, Couloux A, Cruaud C (2009) Evolution of the mitochondrial genome in mammals living at high altitude: new insights from a study of the tribe Caprini (Bovidae, Antilopinae). J Mol Evol 68(4):293–310

Hatcher BG, Johannes RE, Robertson AI (1989) Review of research relevant to the conservation of shallow tropical marine ecosystems. Oceanogr Mar Biol Annu Rev 27:337–414

Hohenlohe PA, Bassham S, Etter PD, Stiffler N, Johnson EA, Cresko WA (2010a) Population genomic analysis of parallel adaptation in threespine stickleback using sequenced RAD tags. PLoS Genet 6:e1000862

Hohenlohe PA, Philips PC, Cresko WA (2010b) Using population genomics to detect selection in natural populations: key concepts and methodological considerations. Int J Plant Sci 171(9):1059–1071. https://doi.org/10.1086/656306

Holthuis LB (1946) XVI On a small collection of isopod crustacea from the greenhouses of the Royal Botanic Gardens, Kew. Annals Magazine Natural History 98(13):122–137. https://doi.org/10.1080/00222934608654533

Holthuis LB, Edwards AJ, Lubbock HR (1980) The decapod and stomatopod crustacea of St Paul's rocks. Zoologische Mededelingen 56(3):27–51

Holthuis LB (1991) FAO species catalogue Vol Marine lobsters of the world An annotated and illustrated catalogue of species of interest to fisheries known to date. FOA Fisheries Synopsis 125(13)

Iftikar FI, MacDonald JR, Baker DW, Renshaw GMC, Hickey AJR (2014) Could thermal sensitivity of mitochondria determine species distribution in a changing climate? J Exp Biol 217:2348–2357. https://doi.org/10.1242/jeb.098798

In von Middendorf (1851) (Decapoda: Caridea: Pandalidae), assembled from linked-reads sequencing. J Crustacean Biol. 42(1):ruac03. https://doi.org/10.1093/jcbiol/ruac003

Jin J-J, Yu W-B, Yang J-B, Song Y, dePamphilis CW, Yi T-S, Li D-Z (2020) GetOrganelle: a fast and versatile toolkit for accurate de novo assembly of organelle genomes. Genome Biol 21:241. https://doi.org/10.1186/s13059-020-02154-5

Jonckheere AI, Hogeveen M, Nijtmans LGJ, van den Brand MAM, Janssen AJM, Diepstra JHS, van den Brandt FCA, van den Heuvel LP, Hol FA, Hofste TGJ, Kapusta L, Dillmann U, Shamdeen MG, Smeitink JAM, Rodenburg RJT (2007) A novel mitochondrial ATP8 gene mutation in a patient with apical hypertrophic cardiomyopathy and neuropathy. J Med Genet 45:129–133. https://doi.org/10.1136/jmg.2007.052084

Jühling F, Pütz J, Bernt M, Donath A, Middendorf M, Florentz C, Stadler PF (2012) Improved systematic tRNA gene annotation allows new insights into the evolution of mitochondrial tRNA structures and into the mechanisms of mitochondrial genome rearrangements. Nucleic Acids Res 40(7):2833–2845. https://doi.org/10.1093/nar/gkr1131

Kerpedjiev P, Hammer S, Hofacker IL (2015) Forna (force-directed RNA): Simple and effective online RNA secondary structure diagrams. Bioinformatics 31(20):3377–3379

Kim G, Yoon T, Park W, Park J, Kang J, Park H, Kim H (2015) Complete mitochondrial genome of Australian spiny lobster, *Panulirus cygnus* (George, 1962) (Crustacea: Decapoda: Palinuridae) from coast of Australia. Mitochondrial DNA 27(6):4576–4577. https://doi.org/10.3109/19401736.2015.1101571

Kumar S, Stecher G, Li M, Knyaz C, Tamura K (2018) MEGA X: molecular evolutionary genetics analysis across computing platforms. Mol Biol Evol 35:1547–1549

Lannig G, Bock C, Sartoris FJ, Pörtner HO (2004) Oxygen limitation of thermal tolerance in cod, *Gadus morhua* L., studied by magnetic resonance imaging and online venous oxygen monitoring. Am J Physiol Regul Intergr Comp Physiol 287:R902–R910

Lavalli KL, Spanier E, Goldstein JS (2019) Scyllarid Lobster Biology and Ecology. In: G. Diarte-Plata, & R. Escamilla-Montes (Eds.), Crustacea. IntechOpen. https://doi.org/10.5772/intechopen.88218

Lee BD (2018) Python implementation of codon adaptation index. J Open Source Softw 3(30):905. https://doi.org/10.21105/joss.00905

Lehninger AL (1964) Molecular basis of structure and function. The Mitochondrion (W.A. Benjamin, New York), 1–4

Li Y, Ren Z, Shedlock AM, Wu J, Sang L, Tersing T, Hasegawa M, Yonezawa T, Zhong Y (2013) High altitude adaptation of the schizothoracine fishes (Cyprinidae) revealed by the mitochondrial genome analyses. Gene 517:169–178. https://doi.org/10.1016/j.gene.2012.12.096

Li T, Yang J, Li Y, Cui Y, Xie Q, Bu W, Hillis DM (2016) A mitochondrial genome of Rhyparochromidae (Hemiptera: Heteroptera) and a comparative analysis of related mitochondrial genomes. Sci Rep 6:351375. https://doi.org/10.1038/srep35175

Liang H (2012) Complete mitochondrial genome of the ornate rock lobster *Panulirus ornatus* (Crustacea: Decapoda). Afr J Biotech 11(80):14519–14528. https://doi.org/10.5897/AJB12.036

Little CTS, Vrijenhoek RC (2003) Are hydrothermal vent animals living fossils? Trends Ecol Evol 18:582–588. https://doi.org/10.1016/j.tree.2003.08.009

Liu Y, Cui Z (2011) Complete mitochondrial genome of the Chinese spiny lobster *Panulirus stimpsoni* (Crustacea: Decapoda): genome characterization and phylogenetic considerations. Mol Biol Rep 38:403–410. https://doi.org/10.1007/s11033-010-0122-2

Liu H, Pu L (2021) The complete mitochondrial genome of longlegged spiny lobster *Panulirus longipes* (A. Milne Edwards 1868). Mitochondrial DNA Part B 6(2):660–662. https://doi.org/10.1080/23802359.2021.1878952

Liu H, Wang G (2020) The complete mitochondrial genome of Aesop slipper lobster *Scyllarides haanii* (De Haan, 1841). Mitochondrial DNA Part B 5(3):3404–3405. https://doi.org/10.1080/23802359.2020.1823274

Liu H, Xia G (2021) The complete mitochondrial genome of the pronghorn spiny lobster *Panulirus pencillatus* (Olivier, 1791). Mitochondrial DNA Part B 6(1):148–150. https://doi.org/10.1080/23802359.2020.1852899

Liu H, Yang M, He Y (2019) Complete mitochondrial genome of blunt slipper lobsters *Scyllarides squammosus* (H Milne Edwards, 1837). Mitochondrial DNA Part B 4(2):2299–2300. https://doi.org/10.1080/23802359.2019.1627935

Liu H, Yang M, He Y (2020) The complete mitochondrial genome of Japanese spear lobster *Linuparus trigonus* (Von Siebold, 1824). Mitochondrial DNA Part B 5(2):1592–1593. https://doi.org/10.1080/23802359.2020.1742618

Luo Y, Yang X, Gao Y (2013) Mitochondrial DNA response to high altitude: A new perspective on high-altitude adaptation. Mitochondrial DNA 24(4):313–319. https://doi.org/10.3109/19401736.2012.760558

Massabuau JC (2001) From low-arterial–to low tissue–oxygenation strategy. Evolutionary Resp Physiol 128:249–261

Michaelsen J, Fago A, Bundgaard A (2021) High temperature impairs mitochondrial function in rainbow trout cardiac mitochondria. J Exp Biol 224(9):jeb242382. https://doi.org/10.1242/jeb.242382

Miller MA, Pfeiffer W, Schwartz T (2010) Creating the CIPRES Science Gateway for inference of large phylogenetic trees. Proceedings of the Gateway Computing Environments Workshop (GCE) 14:1–8

Mishmar D, Ruiz-Pesini E, Golik P, Macaulay V, Clark AG, Hosseini S, Brandon M, Easley K, Chen E, Brown MD (2003) Natural selection shaped regional mtDNA variation in humans. Proc Natl Acad Sci USA 100(1):171–176

Monnens M, Thijs S, Briscoe AG, Clark M, Frost EJ, Littlewood DTJ, Sewell M, Smeets K, Artois T, Vanhoe MPM (2020) The first mitochondrial genomes of endosymbiotic rhabdocoels illustrate evolutionary relaxation of *atp8* and genome plasticity in flatworms. Int J Biol Macromol 162:454–469. https://doi.org/10.1016/j.ijbiomac.2020.06.025

Murrell B, Moola S, Mabona A, Weighill T, Sheward D, Kosakovsky Pond SL, Scheffler K (2013) FUBAR: a fast, unconstrained bayesian approximation for inferring selection. Mol Biol Evol 30(5):1196–1205. https://doi.org/10.1093/molbev/mst030

Murrell B, Weaver S, Smith MD, Wertheim JO, Murrell S, Aylward A, Eren K, Pollner T, Martin DP, Smith DM, Scheffler K, Kosakovsky Pond SL (2015) Gene-wide identification of episodic selection. Mol Biol Evol 32:1365–1371. https://doi.org/10.1093/molbev/msv035

Murrell B, Wertheim JO, Moola S, Weighill T, Scheffler K, Pond SLK, Malik HS (2012) Detecting individual sites subject to episodic diversifying selection. PLOS Genet 8(7):1–10

National Center for Biotechnology Information (NCBI)[Internet]. Bethesda (MD): National Library of Medicine (US), National Center for Biotechnology Information (1988 – 2021). https://www.ncbi.nlm.nih.gov/

Neiman M, Taylor DR (2009) The causes of mutation accumulation in mitochondrial genomes. Proc R Soc B 276:1201–1209. https://doi.org/10.1098/rspb.2008.1758

Nguyen L-T, Schmidt HA, Haeseler AV, Minh BQ (2015) IQ-TREE: a fast and effective stochastic algorithm for estimating maximum likelihood phylogenies. Mol Biol Evol 32:268–274. https://doi.org/10.1093/molbev/msu300

Oellermann M, Hickley AJR, Fitzgibbon QP, Smith G (2020) Thermal sensitivity links to cellular cardiac decline in three spiny lobsters. Nat Sci Rep 10:202. https://doi.org/10.1038/s41598-019-56749-0

Ohmae E, Miyashita Y, Kato C (2013) Thermodynamic and functional characteristics of deep-sea enzymes revealed by pressure effects. Extremophiles 17:701–709. https://doi.org/10.1007/s00792-013-0556-2

Olson-Manning CF, Wagner MR, Mitchell-Olds T (2012) Adaptive evolution: evaluating empirical support for theoretical predictions. Nat Rev Genet 13:867–877. https://doi.org/10.1038/nrg3322

Palero F, Crandall KA, Abelló P, Macpherson E, Pascual M (2009) Phylogenetic relationships between spiny, slipper, and coral lobsters (Crustacea, Decapoda, Achelata). Mol Phylogenet Evol 50:152–162. https://doi.org/10.1016/j.ympev.2008.10.003

Palero F, Genis-Armero R, Hall MR, Clark PF (2016) DNA barcoding the phyllosoma of Scyllarides squammosus (H Milne Edwards, 1837) (Decapoda: Achelata: Scyllaridae). Zootaxa 4139(4):481–498. https://doi.org/10.11646/zootaxa.4139.4.2

Palozzi JM, Jeediguntta SP, Hurd TR (2018) Mitochondrial DNA purifying selection in mammals and invertebrates. J Mol Biol 430(24):4834–4848. https://doi.org/10.1016/j.jmb.2018.10.019

Patek SN, Oakley TH (2003) Comparative tests of evolutionary trade-offs in a Palinurid lobster acoustic system. Evolution 57(9):2082–2100

Pollock DE (1990) Paleoceanography and speciation in the spiny lobster genus Jasus. Bull Mar Sci 46(2):387–405. https://doi.org/10.1016/S0198-0254(06)80473-5

Pond SLK, Frost SDW (2005a) Datamonkey: rapid detection of selective pressure on individual sites of codon alignments. Bioinform Appl Note 21(10):2531–2533. https://doi.org/10.1093/bioinformatics/bti320

Pond SLK, Frost SDW (2005b) Not so different after all: a comparison of methods for detecting amino acid sites under selection. Mol Biol Evol 22(5):1208–1222

Pond SLK, Muse SV (2005) Site-to-site variation of synonymous substitution rates. Mol Biol Evol 22(12):2375–2385

Poon AFY, Frost SDW, Pond SLK (2009) Detecting signatures of selection from DNA sequences using datamonkey. Mol Phylogenet Evol 66:776–789

Price MN, Dehal PS, Arkin AP (2010) FastTree 2 – Approximately Maximum-Likelihood Trees for Large Alignments. PLoS ONE 5:e9490. https://doi.org/10.1371/journal.pone.0009490

Ptacek SN, Sarver SK, Childress MJ, Herrnkind WF (2001) Molecular phylogeny of the spiny lobster genus Panulirus (Decapoda: Palinuridae). Mar Freshwater Res 52:1037–1047. https://doi.org/10.1071/MF01070

Ramos NI, DeLeo DM, Horowitz J, McFadden CS, Quattrini AM (2023) Selection in coral mitogenomes, with insights into adaptations in the deep sea. Nat Sci Rep 13:6016. https://doi.org/10.1038/s41598-023-31243-1

Reuter JS, Mathews DH (2010) RNAstructure: software for RNA secondary structure prediction and analysis. BMC Bioinform 11:129

Rodriguez-Pilco MA, Leśny P, Podsiadłowski L, Schubart CD, Baeza JA (2022) Characterization of the complete mitochondrial genome of the bromeliad crab metopaulias depressus (Rathbun, 1896) (Crustacea: Decapoda: Brachyura: Sesarmidae). Genes 13(2):299. https://doi.org/10.3390/genes13020299

Sabeti PC, Schaffner SF, Fry B, Lohmueller J, Varilly P, Shamovsky O, Palma A, Mikkelsen TS, Altshuler D, Lander ES (2006) Positive natural selection in the human lineage. Science 312:1614–1620. https://doi.org/10.1126/science.1124309

Sambrook J, Fritsch EF, Maniatis T (1989) Molecular cloning: a laboratory manual. Cold Spring Harbor Laboratory Press

Shen H, Braband A, Scholtz G (2013) Mitogenomic analysis of decapod crustacean phylogeny corroborates traditional views on their relationships. Mol Phylogenet Evol 66:776–789. https://doi.org/10.1016/j.ympev.2012.11.002

Sievers F, Higgins DG (2014) Clustal Omega. Curr Protoc Bioinform 48(1):3–13. https://doi.org/10.1002/0471250953.bi0313s48

Sievers F, Higgins DG (2018) Clustal Omega for making accurate alignments of many protein sequences. Protein Sci 27:135–145. https://doi.org/10.1002/pro.3290

Smith MD, Wertheim JO, Weaver S, Murrell B, Scheffler K, Kosakovsky Pond SL (2015) Less is more: an adaptive branch-site random effects model for efficient detection of episodic diversifying selection. Mol Biol Evol 32:1342–1353. https://doi.org/10.1093/molbev/msv022

Stammer D, Martins MS, Köhler J, Köhl A (2021) How well do we know ocean salinity and its changes? Progress Oceanography. https://doi.org/10.1016/j.pocean.2020.102478

Stanley SM (1975) A theory of evolution above the species level. Proc Nat Acad Sci 72(2):6464–6650

Stillman JH (2002) Causes and consequences of thermal tolerance limits in rocky intertidal porcelain crabs. Genus *Petrolisthes* Integ Comp Biol 42:790–796

Stothard P (2000) The sequence manipulation suite: javascript programs for analyzing and formatting protein and DNA sequences. Biotechniques 28:1102–1104

Sun S, Sha Z, Wang Y (2019a) The complete mitochondrial genomes of two vent squat lobsters, *Munidopsis lauensis* and *M. verrilli*: Novel gene arrangements and phylogenetic implications. Ecol Evol 9:12390–12407. https://doi.org/10.1002/ece3.5542

Sun S, Sha Z, Wang Y (2019b) Divergence history and hydrothermal vent adaptation of decapod crustaceans: a mitogenomic perspective. PLoS ONE 14(10):e0224373. https://doi.org/10.1371/journal.pone.0224373

Sun S, Sha Z, Wang Y (2021) Mitochondrial phylogenomics reveal the origin and adaptive evolution of the deep-sea caridean shrimps (Decapoda: Caridea). J Oceanol Limnol 39(5):1948–1960. https://doi.org/10.1007/s00343-020-0266-4

Sun S, Hui M, Wang M, Sha Z (2018) The complete mitochondrial genome of the alvinocaridid shrimp *Shinkaicaris leurokolos* (Decapoda Caridea) Insight into the mitochondrial genetic basis of deep-sea hydrothermal vent adaptation in the shrimp. Comparative Biochemistry and Physiology – Part D 25:42–52. https://doi.org/10.1016/j.cbd.2017.11.002

Tan MH, Gan H, Lee Y, Austin C (2016) The complete mitogenome of the morton bay bug *Thenus orientalis* (Lund, 1793) (Crustacea: Decapoda: Scyllaridae) from a cooked sample and a new mitogenome order for the Decapoda. Mitochondrial DNA Part A 27(2):1277–1278. https://doi.org/10.3109/19401736.2014.945554

Tan MH, Gan HM, Schultz MB, Austin CM (2015) MitoPhAST, a new automated mitogenomic phylogeny tool in the post-genomic era with a case study of 89 decapod mitogenomes including eight new freshwater crayfish mitogenomes. Mol Phylogenet Evol 85:180–188. https://doi.org/10.1016/j.ympev.2015.02.009

Turrens JF (2003) Mitochondrial formation of reactive oxygen species. J Physiol 552:335–344

Ueda T, Yotsumoto Y, Ikeda K, Watanabe K (1992) The T-loop region of animal mitochondrial $tRNA^{Ser}$ (AGY) is a main recognition site for homologous seryl-tRNA synthetase. Nucleic Acids Res 20(9):2217–2222. https://doi.org/10.1093/nar/20.0.2217

Vences M, Miralles A, Brouillet S, Ducasse J, Fedosov A, Kharchev V, Kostadinov I, Kumari S, Patmanidis S, Scherz MD, Puillandre N, Renner SS (2021) Itaxotools 0.1 kickstarting: a specimen-based software toolkit for taxonomists. Megataxa. 6(2):77–92. https://doi.org/10.11646/megataxa.6.2.1

Vences M, Patmanidis S, Kharchev V, Renner SS (2022) Concatenator, a user-friendly program to concatenate DNA sequences, implementing graphical user interfaces for MAFFT and FastTree. Bioinform Adv 2(1):1–4. https://doi.org/10.1093/bioadv/vbac050

Wakeley J (2009) Coalescent theory: an introduction. Roberts, Greenwood Village, CO

Wang Z, Shi X, Sun L, Bai Y, D. Zhang, Tang B (2017) Evolution of mitochondrial energy metabolism genes associated with hydrothermal vent adaptation of Alvinocaridid shrimps. Genes Genom 39:1367–1376. https://doi.org/10.1007/s13258-017-0600-1

Wang D-P, Wan H-L, Zhang S, Yu J (2009) γ-MYN: a new algorithm for estimating Ka and Ks with consideration of variable substitution rates. Biol Direct 4:20

Weaver S, Shank SD, Spielman SJ, Li M, Muse SV, Pond SLK (2018) Datamonkey 2.0: a modern web application for characterizing selection and other evolutionary processes. Mol Biol Evol 35(3):733–777. https://doi.org/10.1093/molbev/msx335

Weiss H, Friedrich T, Hofhaus G, Preis D (1991) The respiratory-chain NADH dehydrogenase (complex I) of mitochondria. Eur J Biochem 197:563–576

Wertheim JO, Murrell B, Smith MD, Pond SLK, Scheffler K (2015) RELAX: Detecting relaxed selection in a phylogenetic framework. Mol Biol Evol 32(3):820–832. https://doi.org/10.1093/molbev/msu400

Xiao B, Zhang W, Yao W, Liu C, Liu L (2017) Analysis of the complete mitochondrial genome sequence of *Panulirus homarus*. Mitochondrial DNA Part B 2(1):60–61. https://doi.org/10.1080/23802359.2017.1280703

Xu X-D, Guan J-Y, Zhang Z-Y, Cao Y-R, Cai Y-Y, Storey KB, Yu D-N, Zhang J-Y (2021) Insight into the phylogenetic relationships among three subfamilies within Heptageniidae (Insecta: Ephemeroptera) along with low-temperature selection pressure analyses using mitogenomes. Insects 12:656. https://doi.org/10.3390/insects12070656

Xu S, Luosang J, Hua S, He J, Ciren A, Wang W, Tong X, Liang Y, Wang J, Zheng X (2007) High altitude adaptation and phylogenetic analysis of tibetan horse based on the mitochondrial genome. J Genet Genomics 34(8):720–729

Yamauchi M, Miya M, Nishida M (2002) Complete mitochondrial DNA sequence of the Japanese spiny lobster, *Panulirus japonicus* (Crustacea: Decapoda). Gene 295:89–96

Yanchukov A, Proulx S (2012) Invasion of gene duplication through masking for maladaptive gene flow. Evolution 66:1543–1555. https://doi.org/10.1111/j.1558-5646.2011.01551.x

Yang M, Gong L, Sui J, Li X (2019a) The complete mitochondrial genome of *Calyptogena marissinica* (Heterodonta: Veneroida: Vesicomyidae): Insight into the deep-sea adaptive evolution of vesicomyids. PLoS ONE 14(9):e0217952

Yang C, Liu Y, Cui Z, Chan T (2014) Complete mitochondrial genome of the furry lobster *Palinurellus wieneckii* (De Man, 1881) (Decapoda, Achelata, Palinuridae). Mitochondrial DNA 25(4):295–297. https://doi.org/10.3109/19401736.2013.796521

Yang M, Liu H, He Y (2019b) Complete mitochondrial genome of sculptured slipper lobster *Parribacus antarcticus* (Lund, 1793). Mitochondrial DNA Part B 4(2):2297–2298. https://doi.org/10.1080/23802359.2019.1627934

Yu L, Wang X, Ting N, Zhang Y (2011) Mitogenomic analysis of Chinese snub-nosed monkeys: evidence of positive selection in NADH dehydrogenase gene in high-altitude adaptation. Mitochondrion 11:497–503. https://doi.org/10.1016/j.mito.2011.01.004

Zhang D, Gong F, Liu T, Guo H, Zhang N, Zhu K, Jiang S (2015) Shotgun assembly of the mitochondrial genome from Fenneropenaeus penicillatus with phylogenetic consideration. Mar Genomics 24:379–386. https://doi.org/10.1016/j.margen.2015.09.005

Zhang HX, QiBin L, Jing S, Fei L, Gang W, Jun Y, WeiWei W (2013) Mitochondrial genome sequences of *Artemia tibetiana* and *Artemia urmiana*: assessing molecular changes for high plateau adaptation. Sci China Life Sci 56(5):440–452. https://doi.org/10.1007/s11427-013-4474-4

Zhou T, Shen X, Irwin DM, Shen Y, Zhang Y (2014) Mitogenomic analyses propose positive selection in mitochondrial genes for high-altitude adaptation in galliform birds. Mitochondrion 18:70–75. https://doi.org/10.1016/j.mito.2014.07.012

joseba (2012) BioPHP - Microsatellite Repeats Finder.